

QUANTITATIVE FINANCE

PART I - MEAN-VARIANCE ANALYSIS

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What is important is the gradual development of a theory, based on a careful analysis of the ... facts.

- ... Its **first** applications are necessarily to elementary problems where the result has never been in doubt and no theory is actually required. At this early stage the application serves to corroborate the theory.
- The **next** stage develops when the theory is applied to somewhat more complicated situations in which it may already lead to a certain extent beyond the obvious and familiar. Here theory and application corroborate each other mutually.
- **Beyond** lies the field of real success: genuine prediction by theory.

It is well known that all mathematized sciences have gone through these successive stages of evolution.

—— John von Neumann

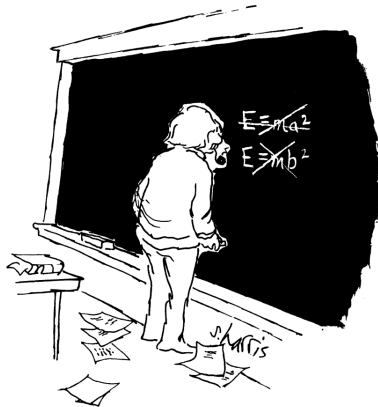
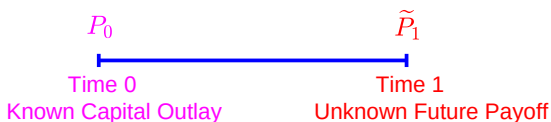


Figure 4.4. How did Einstein derive his famous formula? (Courtesy Sidney Harris)

(Source: Anthony Zee, 1986. *Fearful symmetry: The search for beauty in modern physics*. Princeton University Press.)

The Single Period Setup



Two most important things in asset pricing:

- 1 Time value
- 2 Uncertainty

When is a single period model a good approximation?

- 1 A zero coupon bond held to maturity
- 2 A physical project providing no dividends until completed

Returns



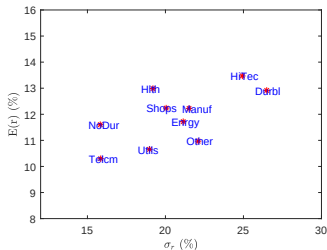
If we have a long position, then

- Initial outlay is X_0 . Final receipt is X_1 .
- The total return is $R = X_1/X_0$.

If we **short sell** the asset, then

- Initial outlay is $-X_0$. Final receipt is $-X_1$.
- The total return is $R = (-X_1)/(-X_0) = X_1/X_0$.

Mean-Standard Deviation Diagram



- 1 NoDur Consumer Nondurables -- Food, Tobacco, Textiles, Apparel, Leather, Toys
- 2 Durbl Consumer Durables -- Cars, TVs, Furniture, Household Appliances
- 3 Manuf Manufacturing -- Machinery, Trucks, Planes, Chemicals, Off Furn, Paper, Com Printing
- 4 Enrgy Oil, Gas, and Coal Extraction and Products
- 5 HlTec Business Equipment -- Computers, Software, and Electronic Equipment
- 6 Telcm Telephone and Television Transmission
- 7 Shops Wholesale, Retail, and Some Services (Laundries, Repair Shops)
- 8 Hlth Healthcare, Medical Equipment, and Drugs
- 9 Utlis Utilities
- 10 Other Other -- Mines, Constr, BldMt, Trans, Hotels, Bus Serv, Entertainment, Finance

A simple method to **represent randomness** in returns is through a two-dimensional mean-standard deviation diagram. (Why not mean-variance?)

The above figure shows the 10 industry portfolios of the US equity market. Here we use monthly returns from July 1926 to January 2020 to do the calculation. We report means and standard deviations on an annualized basis.

Recap

We invest a total amount of X_0 in N assets, with amount X_{0i} for asset i , $i = 1, 2, \dots, N$. Let $w_i = X_{0i}/X_0$ be the weight of asset i . Obviously, $\sum_{i=1}^N w_i = 1$.

Then, the **portfolio return** is

$$R_p = \frac{\sum_{i=1}^N R_i w_i X_0}{X_0} = \sum_{i=1}^N w_i R_i.$$

Equivalently,

$$r_p = \sum_{i=1}^N w_i r_i, \tag{1}$$

because $\sum_{i=1}^N w_i = 1$.

Portfolio Mean and Variance

Let $\bar{r}_i = E(r_i)$.

Write $\bar{\mathbf{r}} = (\bar{r}_1, \bar{r}_2, \dots, \bar{r}_N)^\top$ and $\mathbf{w} = (w_1, w_2, \dots, w_N)^\top$.

Let $\Sigma = [\sigma_{ij}]$ be the $N \times N$ variance-covariance matrix. Here, $\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$, where ρ_{ij} is the correlation coefficient between asset i 's and j 's returns, and σ_i and σ_j are the standard deviation of asset i 's and j 's returns, respectively.

Then, the **portfolio mean and variance** are

$$\bar{r}_p = \sum_{i=1}^N w_i \bar{r}_i = \mathbf{w}^\top \bar{\mathbf{r}}, \quad (2)$$

$$\sigma_p^2 = E[(r_p - \bar{r}_p)^2] = \sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{ij} = \mathbf{w}^\top \Sigma \mathbf{w}. \quad (3)$$

Notice that all **vectors and matrices** are shown in **bold**.

Exercise 1

Calculate the mean and variance of an equally weight portfolio on the 10 industry portfolios. Use monthly returns provided in *10_Industry_Portfolios.xlsx*. (First estimate the means and covariances for the 10 industry portfolios.)

Excel provides built-in functions for matrix manipulations.

- ***MMULT(A, B)*** performs matrix multiplications;
- ***TRANSPOSE(A)*** yields a matrix transpose;
- ***MUNIT(N)*** generates a $N \times N$ identity matrix;
- ***A*B*** gives the element by element multiplication;
- ***A*MUNIT(N)*** returns the diagonal matrix of ***A***.

Press **Ctrl+Shift+Enter**, instead of Enter, to get results involving matrix functions.

The Markowitz Model

Put forth by Markowitz (1952).¹

There are N assets with expected returns $\bar{r}_1, \bar{r}_2, \dots, \bar{r}_N$, and covariances $\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$, $i, j = 1, 2, \dots, N$.

A portfolio formed with the above assets is defined by the weights $w_1, w_2, \dots, w_N$ on each individual asset.

The quest is to find the minimum-variance portfolio for any (feasible) desired level of expected return.

¹Markowitz, Harry M., 1952. Portfolio Selection, *Journal of Finance* 7, 77-91.

Mathematical Optimization

An (mathematical) *optimization problem* has the form

$$\begin{aligned} & \underset{\mathbf{x}}{\text{minimize}} && f(\mathbf{x}), \\ & \text{subject to} && \mathbf{g}(\mathbf{x}) \leq \mathbf{c}, \end{aligned} \tag{4}$$

where the vector $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$ is the *optimization variable*, the function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the *objective function*, the function $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the (equality or inequality) *constraint function*, and the constant vector $\mathbf{c} \in \mathbb{R}^m$ is the *bound* for the constraint.

A maximization problem needs no separate treatment because it can be easily transformed into an equivalent minimization problem by simply putting a negative sign before the objective.

$$\underset{\mathbf{x}}{\text{maximize}} \quad f(\mathbf{x}) \quad \Leftrightarrow \quad \underset{\mathbf{x}}{\text{minimize}} \quad -f(\mathbf{x}). \tag{5}$$

A Simplified Version for Mean-Variance Analysis

We simplify notation and also the *optimization problem* as follows.

$$\begin{array}{ll} \min_{\mathbf{x}} & f(\mathbf{x}), \\ \text{s.t.} & g_i(\mathbf{x}) = c_i, \quad i = 1, 2, \dots, m, \end{array} \quad (6)$$

where the vector $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$ is the *optimization variable*, the function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the *objective function*, the functions $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $i = 1, 2, \dots, m$, are the *equality constraint functions*, and the constants c_i , $i = 1, 2, \dots, m$, are the *bounds* for the constraints.

We will have more discussions on inequality constraints and mean-variance analysis after we finish this simplified case.

Formulating the Markowitz Model

Fix the expected return of a portfolio at $\bar{r}_p$. We want to find the minimum-variance portfolio that achieves $\bar{r}_p$.

$$\begin{aligned} \min_{w_1, w_2, \dots, w_N} \quad & \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N w_i \sigma_{ij} w_j, \\ \text{s.t.} \quad & \sum_{i=1}^N w_i \bar{r}_i = \bar{r}_p, \\ & \sum_{i=1}^N w_i = 1. \end{aligned} \tag{7}$$

The factor $\frac{1}{2}$ is innocuous and just works to make the solution neater.

Solution

The Markowitz Model provides the **foundation for single-period investment decisions** by explicitly addressing the **tradeoff between rewards and risk** of a portfolio. He breaks the ice by giving rewards and risk **easily measurable contents**. In particular, he represents rewards by mean return and risk by variance of return. His measures, even if disputable, are admittedly less plagued by subjective judgements.

We solve the model using the **Lagrangian method**. We form the **Lagrangian function**

$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N w_i \sigma_{ij} w_j - \lambda \left(\sum_{i=1}^N w_i \bar{r}_i - \bar{r}_p \right) - \zeta \left(\sum_{i=1}^N w_i - 1 \right), \quad (8)$$

where λ and ζ are the so-called **Lagrangian Multipliers**.

Solution - cont'd

Differentiating the Lagrangian w.r.t. the weights and multipliers, we get the following first order conditions (F.O.C.s).

$$\sum_{j=1}^N \sigma_{ij} w_j - \lambda \bar{r}_i - \zeta = 0, \text{ for } i = 1, 2, \dots, N, \quad (9)$$

$$\sum_{i=1}^N w_i \bar{r}_i = \bar{r}_p, \quad (10)$$

$$\sum_{i=1}^N w_i = 1, \quad (11)$$

where we used the fact $\sigma_{ij} = \sigma_{ji}$ in (9).

We have $N + 2$ linear equations for $N + 2$ unknowns. We can in principle solve the model with linear algebra methods.

A Simple Case with Two Assets

Before presenting the general solution, we first consider the case of two individual assets with expected returns $\bar{r}_1$ and $\bar{r}_2$ ($\bar{r}_1 \neq \bar{r}_2$), and covariances σ_1^2 , σ_{12} and σ_2^2 . The F.O.C.s are:

$$w_1\sigma_1^2 + w_2\sigma_{12} - \lambda\bar{r}_1 - \zeta = 0,$$

$$w_1\sigma_{12} + w_2\sigma_2^2 - \lambda\bar{r}_2 - \zeta = 0,$$

$$w_1\bar{r}_1 + w_2\bar{r}_2 = \bar{r}_p,$$

$$w_1 + w_2 = 1.$$

The last two equations give

$$w_1^* = \frac{\bar{r}_p - \bar{r}_2}{\bar{r}_1 - \bar{r}_2}, \quad w_2^* = \frac{\bar{r}_1 - \bar{r}_p}{\bar{r}_1 - \bar{r}_2}.$$

An Illustration

For the simple case of two assets, only one combination of the assets achieves the required portfolio expected return $\bar{r}_p$. The minimization problem degenerates.

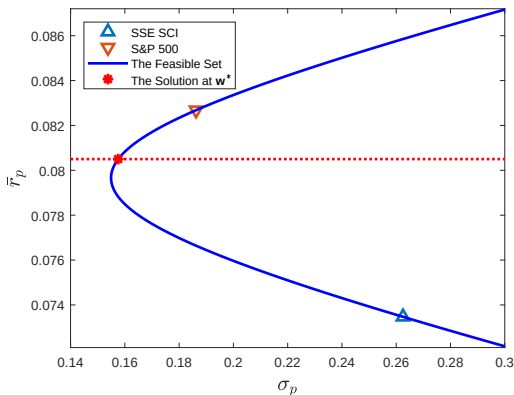
We use the SSE Stock Composite Index and S&P 500 Index as two aggregate stocks (or ETFs), for illustration.

	$\bar{r}_i$	σ_i	ρ
SSE SCI	7.347%	26.25%	4.27%
S&P 500	8.268%	18.63%	

All quantities are annualized when possible.

The Feasible Set

The Feasible Set: The set of portfolios that deliver the targeted expected returns (i.e., satisfy the corresponding constraints), but do not necessarily cater to minimum variance.



A Case of Three Uncorrelated Assets

Now consider a slightly more complicated case of three **uncorrelated** assets ($\sigma_{ij} = 0, i \neq j.$), with expected returns $\bar{r}_1, \bar{r}_2$ and $\bar{r}_3$, and the **same** variance σ^2 . The F.O.C.s are:

$$w_1\sigma^2 - \lambda\bar{r}_1 - \zeta = 0,$$

$$w_2\sigma^2 - \lambda\bar{r}_2 - \zeta = 0,$$

$$w_3\sigma^2 - \lambda\bar{r}_3 - \zeta = 0,$$

$$w_1\bar{r}_1 + w_2\bar{r}_2 + w_3\bar{r}_3 = \bar{r}_p,$$

$$w_1 + w_2 + w_3 = 1.$$

We could **no longer** solve for the w_i s from the last two equations.

A Further Simplification

Let $\bar{r}_1 = 1$, $\bar{r}_2 = 2$ and $\bar{r}_3 = 3$, and $\sigma^2 = 1$. The F.O.C.s are:

$$w_1 - \lambda - \zeta = 0, \quad (12)$$

$$w_2 - 2\lambda - \zeta = 0, \quad (13)$$

$$w_3 - 3\lambda - \zeta = 0, \quad (14)$$

$$w_1 + 2w_2 + 3w_3 = \bar{r}_p, \quad (15)$$

$$w_1 + w_2 + w_3 = 1. \quad (16)$$

Solution

Now, $(12)+(14)-2\times(13)$, we obtain:

$$w_1 - 2w_2 + w_3 = 0, \quad (17)$$

$$w_1 + 2w_2 + 3w_3 = \bar{r}_p, \quad (18)$$

$$w_1 + w_2 + w_3 = 1. \quad (19)$$

Then, $(19)-(17)$ gives

$$w_2^* = 1/3.$$

And

$$w_1^* = -\bar{r}_p/2 + 4/3,$$

$$w_3^* = \bar{r}_p/2 - 2/3.$$

Discussion on the Effect of $\bar{r}_p$ and $\bar{r}_i$

We can write the solution as:

$$w_2^* = 1/3, \quad (20)$$

$$w_1^* = -\bar{r}_p/2 + 4/3 = 1/3 - (\bar{r}_p/2 - 1), \quad (21)$$

$$w_3^* = \bar{r}_p/2 - 2/3 = 1/3 + (\bar{r}_p/2 - 1). \quad (22)$$

Note the interesting special case of $\bar{r}_p = 2 (= (\bar{r}_1 + \bar{r}_2 + \bar{r}_3)/3)$, which results in $w_1^* = w_2^* = w_3^* = 1/3$.

- A higher $\bar{r}_p (> 2)$ increases investment in Asset 3 while decreases investment in Asset 1.
- A lower $\bar{r}_p (< 2)$ decreases investment in Asset 3 while increases investment in Asset 1.
- The level of $\bar{r}_p$ does not affect investment in Asset 2 because of the symmetry in $\bar{r}_i$. No shorting is needed if $4/3 \leq \bar{r}_p \leq 8/3$.

An Illustration

The minimum variance at $\bar{r}_p$ is

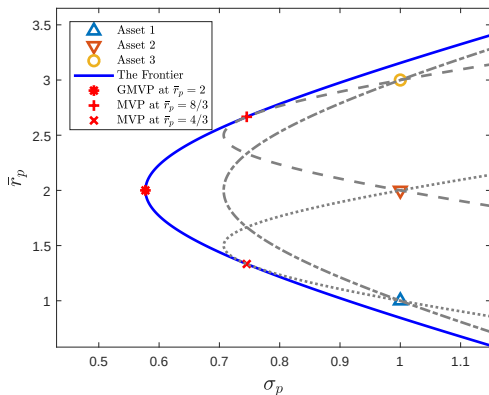
$$\sigma_p^2 = w_1^{*2} + w_2^{*2} + w_3^{*2} = \frac{\bar{r}_p^2}{2} - 2\bar{r}_p + \frac{7}{3}.$$

Let $\frac{\partial \sigma_p^2}{\partial \bar{r}_p} = 0$. We compute the Global Minimum Variance Portfolio (GMVP) at :

$$\begin{aligned}\bar{r}_p^G &= 2, \\ \sigma_p^G &= \frac{\sqrt{3}}{3}, \\ w_1^G &= w_2^G = w_3^G = \frac{1}{3}.\end{aligned}$$

The Efficient Frontier

The Efficient Frontier: The set of minimum variance portfolios from solving the optimization problem for all possible levels of $\bar{r}_p$.



The Maximum Variance

We ask the opposite question: What is the **maximum** variance of a portfolio with expected return $\bar{r}_p$?

- 1 For the case of two assets, any $\bar{r}_p$ is paired with a single σ_p .
- 2 For the case of three assets, we can show that, from (18) and (19),

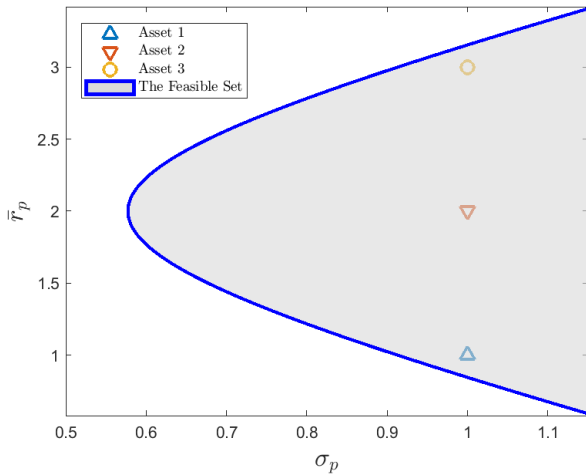
$$w_2 = -2w_1 - \bar{r}_p + 3$$

$$w_3 = w_1 + \bar{r}_p - 2,$$

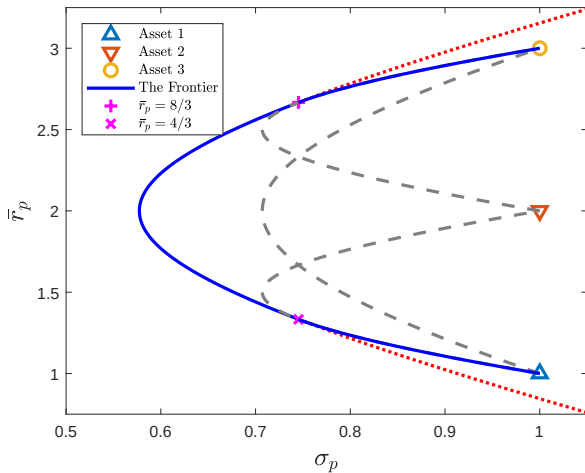
$$\sigma_p^2 = w_1^2 + w_2^2 + w_3^2 = 6w_1^2 + 2(3\bar{r}_p - 8)w_1 + 2\bar{r}_p^2 - 10\bar{r}_p + 13.$$

Obviously, σ_p^2 is **unbounded** if short selling is allowed. That is, σ_p^2 could be any value in $[\frac{1}{2}\bar{r}_p^2 - 2\bar{r}_p + \frac{7}{3}, \infty)$, at any fixed $\bar{r}_p$.

The Feasible Set



The Efficient Frontier under Short-Sale Constraint



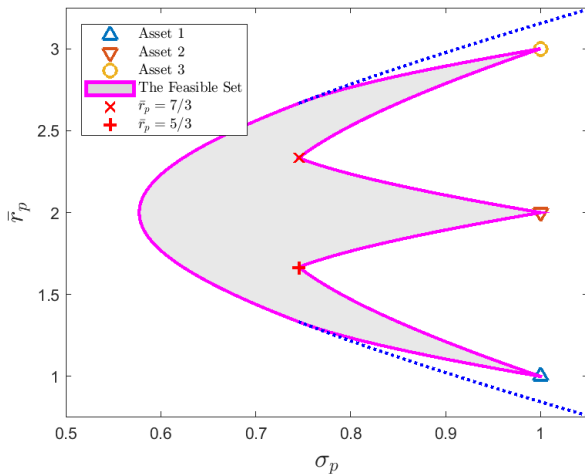
The Efficient Frontier under Short-Sale Constraint

For the simple case of three uncorrelated assets above, we can find the efficient frontier explicitly through combining the pictorial analysis on the previous slide and the solutions in (20)-(22), **when shorting is not allowed**.

	$1 \leq \bar{r}_p < 4/3$	$4/3 \leq \bar{r}_p \leq 8/3$	$8/3 < \bar{r}_p \leq 3$
w_1	$2 - \bar{r}_p$	$1/3 - (\bar{r}_p/2 - 1)$	0
w_2	$\bar{r}_p - 1$	$1/3$	$3 - \bar{r}_p$
w_3	0	$1/3 + (\bar{r}_p/2 - 1)$	$\bar{r}_p - 2$
σ_p	$\sqrt{2\bar{r}_p^2 - 6\bar{r}_p + 5}$	$\sqrt{\bar{r}_p^2/2 - 2\bar{r}_p + 7/3}$	$\sqrt{2\bar{r}_p^2 - 10\bar{r}_p + 13}$
No Shorting	Binding	Not Binding	Binding

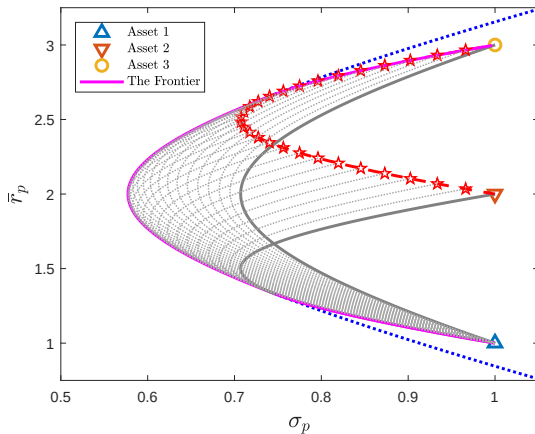
Note that under short-sale constraint, we are **unable** to form a portfolio with $\bar{r}_p < 1$ or $\bar{r}_p > 3$. (**Why?**)

The Feasible Set under Short-Sale Constraint



How to Deal with Shorting Constraints?

The trick is to watch the area swept by the curve!



A “Very” Brief History of Optimization

Most of the theoretical and computational interest in optimization, **nonlinear** in particular, has taken place since 1947.

However, it is useful to note that nonlinear optimization was studied as early as the 1600s. Indeed, both Fermat, in 1638, and Newton, in 1670, studied the **1-dimensional nonlinear unconstrained** optimization problem:

$$\min_{x \in \mathbb{R}} f(x), \quad (23)$$

and Newton developed the classical optimality condition: If f is differentiable at a minimizer x^* ,

$$\frac{df(x^*)}{dx} = 0.$$

A “Very” Brief History of Optimization - Cont'd

This was generalized by Euler (1755) to **multivariate nonlinear unconstrained** optimization:

$$\min_{\mathbf{x}=(x_1, x_2, \dots, x_n)^\top \in \mathbb{R}^n} f(\mathbf{x}), \quad (24)$$

with the optimality condition: If f is differentiable at a minimizer $\mathbf{x}^*$,

$$\nabla f(\mathbf{x}^*) = \mathbf{0},$$

where $\nabla = \frac{\partial}{\partial \mathbf{x}}$ is the Nabla operator, read “del”. We call $\nabla f(\mathbf{x}) = \left(\frac{\partial f(\mathbf{x})}{\partial x_1}, \frac{\partial f(\mathbf{x})}{\partial x_2}, \dots, \frac{\partial f(\mathbf{x})}{\partial x_n} \right)^\top$ the gradient of f .

A “Very” Brief History of Optimization - Cont'd

In the late 1700s, both Euler and Lagrange examined problems in mechanics and established **equilibrium** conditions, as well as generalizations to **infinite** dimensions, developing the **calculus of variations**.

The study of optimization received little attention thereafter until 1947, when **modern linear optimization** was developed by Dantzig,² with an immediate impact, particularly in management.

²A classical reference on the subject is Dantzig (1963): *Linear Programming and Extensions*, Princeton University Press.

A “Very” Brief History of Optimization - Cont'd

In the late 1940s and early 1950s, Kuhn and Tucker (preceded by Karush) developed general optimality conditions for **nonlinear constrained optimization**. At that time, applications of nonlinear optimization included problems in chemical engineering, portfolio optimization, and a host of decision problems in management and industrial operations.

A “Very” Brief History of Optimization - Cont'd

The 1960s saw the development of **larger scale** nonlinear optimization coincident with the expanding capabilities of computers.

In 1984, new possibilities in optimization methods were developed with Karmarkar's **interior-point method** for **linear optimization**, which was extended to **nonlinear convex optimization** by Nesterov and Nemirovski in 1994.

In the 1990s, the important fields of **semidefinite optimization**, **conic optimization**, and “**robust optimization**” were developed.

A “Very” Brief History of Optimization - Cont'd

Since 2000, there has been great interest and progress in “huge-scale” nonlinear optimization, which considers problems in thousands or even millions of decision variables and constraints. Many of these problems have appeared only in recent years, pushed by the Data Science revolution, and other come from domains like astronomical observations.

The above discussion is adapted from Freund and Vera (2019): *Fundamentals of Nonlinear Optimization: a Constructive Approach*, In Preparation, and Cornuéjols, Peña, and Tütüncü (2018): *Optimization Methods in Finance*, Cambridge University Press.

A Trichotomy of Optimization Problems

Let $\mathcal{X}$ denote a subset of $\mathbb{R}^n$. Generally, we can write an *optimization problem* as follows.

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}), \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{X}. \end{aligned} \tag{25}$$

- 1 Convex Optimization:** The objective $f(\mathbf{x})$ is a convex function and the constraint set $\mathcal{X}$ is a convex set. An example is as follows.

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}), \\ \text{s.t.} \quad & g_i(\mathbf{x}) \leq 0, i = 1, 2, \dots, m, \\ & h_j(\mathbf{x}) = 0, j = 1, 2, \dots, n, \end{aligned} \tag{26}$$

where $f(\mathbf{x})$ and $g_i(\mathbf{x})$ are convex functions, and $h_j(\mathbf{x})$ are affine functions.

The class of optimization models is **tractable most of the time**. Hence if you formulate your problem into this class, you can expect it to be amenable to an efficient algorithm. Fortunately, mean-variance analysis anchors here. The maximization of concave utility functions in Part III also fits into this class after proper reformulations.

A Trichotomy of Optimization Problems

Let $\mathcal{X}$ denote a subset of $\mathbb{R}^n$. Generally, we can write an *optimization problem* as follows.

$$\begin{array}{ll} \min_{\mathbf{x}} & f(\mathbf{x}), \\ \text{s.t.} & \mathbf{x} \in \mathcal{X}. \end{array} \quad (27)$$

- 2 **Nonconvex Optimization:** The objective or constraint functions are nonlinear, but not known to be convex. **This nonconvexity poses further challenge for tractability.** We can use the Lagrangian Method in consort with bordered Hessian tests for these problems. But due to limited applications in this course, we leave this excitement for future pursuit.
- 3 **Stochastic and dynamic optimization:** This class of optimization models involves random and time-dependent features, and hence is **tractable only in some special cases**. Unless some specific structure and assumptions hold, a model of the class would typically be insoluble with any realistic amount of computational power at disposal. Current research is striving to enrich tractable models in the class. We will encounter them in Part IV.

The Plan

Before discussing convex optimization in detail, we will first refresh the definitions and properties of **convex sets and functions**. Then we will explain optimization methods starting from the simplest case of univariate unconstrained minimization. Finally, we will formulate mean-variance analysis as a special case of convex optimization, in particular, **quadratic optimization**, also known as **quadratic programming**.

Reassure ourselves!

A quote from Boyd and Vandenberghe (2004)

*“With only a bit of exaggeration, we can say that, if you **formulate** a practical problem as a convex optimization problem, then you **have solved** the original problem.”*

(Boyd, Stephen and Lieven Vandenberghe, 2004. *Convex Optimization*, Cambridge University Press. Excerpt from Section 1.3.2.)

Lines and Line Segments

Suppose $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$. The *line* passing through $\mathbf{x}_1$ and $\mathbf{x}_2$ consists of points of the form

$$\mathbf{y} = \theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2 = \mathbf{x}_2 + \theta(\mathbf{x}_1 - \mathbf{x}_2), \quad (28)$$

where $\theta \in \mathbb{R}$. The *line segment* between $\mathbf{x}_1$ and $\mathbf{x}_2$ corresponds to $\theta \in [0, 1]$.

Affine Sets

A set $\mathcal{A} \subseteq \mathbb{R}^n$ is *affine* if the line through **any two points** in $\mathcal{A}$ lies in $\mathcal{A}$, i.e., if for any $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{A}$, and $\theta \in \mathbb{R}$, we have $\theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2 \in \mathcal{A}$.

This idea can be generalized to more than two points. If $\mathcal{A}$ is an affine set, $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k \in \mathcal{A}$, and $\theta_1 + \theta_2 + \dots + \theta_k = 1$, then the point $\theta_1 \mathbf{x}_1 + \theta_2 \mathbf{x}_2 + \dots + \theta_k \mathbf{x}_k$ also belongs to $\mathcal{A}$. Prove by induction.

Proof.

The case $k = 1$ is trivial.

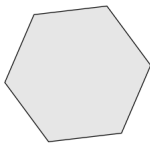
- i) The case $k = 2$ is the definition itself.
- ii) Suppose the case $k = n > 2$ obtains. Then $\theta_1 \mathbf{x}_1 + \theta_2 \mathbf{x}_2 + \dots + \theta_n \mathbf{x}_n \in \mathcal{A}$.
- iii) For the case $k = n + 1$, since $\theta_1 + \theta_2 + \dots + \theta_{n+1} = 1$, there is at least one $\theta_i \neq 1$, $i \in \{1, 2, \dots, n + 1\}$. W.l.o.g, take $\theta_{n+1} \neq 1$. Then $\theta_1 \mathbf{x}_1 + \theta_2 \mathbf{x}_2 + \dots + \theta_{n+1} \mathbf{x}_{n+1} =$
 $(1 - \theta_{n+1}) \underbrace{(\vartheta_1 \mathbf{x}_1 + \vartheta_2 \mathbf{x}_2 + \dots + \vartheta_n \mathbf{x}_n)}_{\in \mathcal{A}} + \theta_{n+1} \mathbf{x}_{n+1} \in \mathcal{A}$, where

$$\vartheta_i = \frac{\theta_i}{1 - \theta_{n+1}} = \frac{\theta_i}{\theta_1 + \theta_2 + \dots + \theta_n}, i = 1, 2, \dots, n, \text{ and } \vartheta_1 + \vartheta_2 + \dots + \vartheta_n = 1.$$

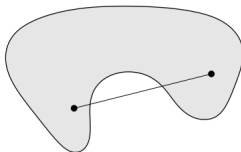


Convex Sets

A set $\mathcal{C} \subseteq \mathbb{R}^n$ is **convex** if the line segment between **any two points** in $\mathcal{C}$ lies in $\mathcal{C}$, i.e., if for any $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{C}$ and $\theta \in [0, 1]$, we have $\theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2 \in \mathcal{C}$. Obviously, **every affine set is also convex**.



Left: Convex



Middle: Nonconvex



Right: Nonconvex

We call a point of the form $\theta_1 \mathbf{x}_1 + \theta_2 \mathbf{x}_2 + \cdots + \theta_k \mathbf{x}_k$, where $\theta_i \geq 0$, $i = 1, 2, \dots, k$, and $\theta_1 + \theta_2 + \cdots + \theta_k = 1$, a convex combination of the points $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$. As with affine sets, it can be shown that a set is convex **if and only if** it contains every convex combination of its points. A **convex combination** of points can be thought of as a mixture or weighted average of the points, with θ_i the fraction of $\mathbf{x}_i$ in the mixture.

Proof by Induction*

Proof.

The “if” part is obvious. Just take $k = 2$.

Now we prove the “only if” part.

- i) The case $k = 1$ is trivial and the case $k = 2$ is the definition itself.
- ii) Suppose the case $k = n > 2$ obtains. Then $\theta_1 \mathbf{x}_1 + \theta_2 \mathbf{x}_2 + \cdots + \theta_n \mathbf{x}_n \in \mathcal{C}$, $\forall \mathbf{x}_i \in \mathcal{C}$, $\theta_i \geq 0$, $i = 1, 2, \dots, n$, and $\theta_1 + \theta_2 + \cdots + \theta_n = 1$.
- iii) For the case $k = n + 1$, given $\theta_1 + \theta_2 + \cdots + \theta_{n+1} = 1$ and $\theta_i \geq 0$, $i = 1, 2, \dots, n + 1$, there is at least one $\theta_i < 1$, $i \in \{1, 2, \dots, n + 1\}$.
W.l.o.g, take $\theta_{n+1} < 1$. Then $\theta_1 \mathbf{x}_1 + \theta_2 \mathbf{x}_2 + \cdots + \theta_{n+1} \mathbf{x}_{n+1} = (1 - \theta_{n+1}) \underbrace{(\vartheta_1 \mathbf{x}_1 + \vartheta_2 \mathbf{x}_2 + \cdots + \vartheta_n \mathbf{x}_n)}_{\in \mathcal{C}} + \theta_{n+1} \mathbf{x}_{n+1} \in \mathcal{C}$, where

$$\vartheta_i = \frac{\theta_i}{1 - \theta_{n+1}} = \frac{\theta_i}{\theta_1 + \theta_2 + \cdots + \theta_n} \geq 0, i = 1, 2, \dots, n, \text{ and } \vartheta_1 + \vartheta_2 + \cdots + \vartheta_n = 1.$$



Important Examples

- The empty set $\emptyset$, any single point (i.e., singleton) $\mathbf{x}_0$, and the whole space $\mathbb{R}^n$ are affine (hence, convex) subsets of $\mathbb{R}^n$.
- Any line is affine (hence, convex). A line segment is convex, but not affine (unless it reduces to a point).
- A *hyperplane* is a set $\mathcal{H} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}^\top \mathbf{x} = b\}$, where $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{a} \neq \mathbf{0}$, and $b \in \mathbb{R}$. For any $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{H}$, let $\mathbf{y} = \theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2$, where $\theta \in \mathbb{R}$. We have $\mathbf{a}^\top \mathbf{y} = \theta \mathbf{a}^\top \mathbf{x}_1 + (1 - \theta) \mathbf{a}^\top \mathbf{x}_2 = \theta b + (1 - \theta)b = b$. Thus $\mathbf{y} \in \mathcal{H}$. This proves $\mathcal{H}$ is an affine (hence, convex) set.

Question 1

Show the following result.

Intersection of Affine (Hence, Convex) Sets

If $\mathcal{C}_1, \mathcal{C}_2 \subseteq \mathbb{R}^n$ are two affine (hence, convex) sets, then $\mathcal{C}_1 \cap \mathcal{C}_2$ is also affine (hence, convex).

Question 1 - Answer

Intersection of Affine (Hence, Convex) Sets

If $\mathcal{C}_1, \mathcal{C}_2 \subseteq \mathbb{R}^n$ are two affine (hence, convex) sets, then $\mathcal{C}_1 \cap \mathcal{C}_2$ is also affine (hence, convex).

If $\mathcal{C}_1 \cap \mathcal{C}_2$ is empty, the result is obvious.

Otherwise, for any $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{C}_1 \cap \mathcal{C}_2$, clearly, $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{C}_1$ and $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{C}_2$. Let $\mathbf{y} = \theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2$, where $\theta \in \mathbb{R}$. Since both $\mathcal{C}_1$ and $\mathcal{C}_2$ are affine, $\mathbf{y} \in \mathcal{C}_1$ and $\mathbf{y} \in \mathcal{C}_2$. Thus $\mathbf{y} \in \mathcal{C}_1 \cap \mathcal{C}_2$.

We can extend the result to intersection of any (possibly infinite) number of sets.

Question 1 - Extended*

Intersection of Countable Affine (Hence, Convex) Sets

If $\mathcal{C}_i \subseteq \mathbb{R}^n$, $i = 1, 2, \dots$, is each an affine (hence, convex) set, then $\mathcal{C} = \cap_{i=1}^{\infty} \mathcal{C}_i$ is also affine (hence, convex).

If $\mathcal{C}$ is empty, the result is obvious.

Otherwise, for any $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{C}$, clearly, $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{C}_i$, $i = 1, 2, \dots$. Let $\mathbf{y} = \theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2$, where $\theta \in \mathbb{R}$. Since all $\mathcal{C}_i$ s are affine, $\mathbf{y} \in \mathcal{C}_i$, $i = 1, 2, \dots$. Thus $\mathbf{y} \in \mathcal{C}$.

For the case of intersection of a finite number of affine sets, say, N sets, take affine sets $\mathcal{C}_{N+i} = \mathbb{R}^n$, $i = 1, 2, \dots$. Clearly, $\cap_{i=1}^N \mathcal{C}_i = \cap_{i=1}^{\infty} \mathcal{C}_i$ and the latter is affine using the above result.

Important Examples - Cont'd

- The *solution set* to a system of linear equations is $\mathcal{S} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{A}^\top \mathbf{x} = \mathbf{b}\}$, where $\mathbf{A} \in \mathbb{R}^{n \times m}$, $\mathbf{A} \neq \mathbf{0}$, and $\mathbf{b} \in \mathbb{R}^m$. Note that $\mathcal{S}$ is the intersection of m hyperplanes ($\mathbb{R}^n$ is also a hyperplane) and thus an affine (hence, convex) set (possibly an empty set).
- A *Euclidean ball*, or just *ball*, in $\mathbb{R}^n$ is $B_r(\mathbf{x}_0) = \{\mathbf{x} \mid \|\mathbf{x} - \mathbf{x}_0\| < r\} = \{\mathbf{x} \mid (\mathbf{x} - \mathbf{x}_0)^\top (\mathbf{x} - \mathbf{x}_0) < r^2\}$, where $r > 0$ and $\|\mathbf{u}\| = (\mathbf{u}^\top \mathbf{u})^{1/2}$ denotes Euclidean norm. The vector $\mathbf{x}_0$ is the center of the ball and the scalar r is its radius.
A ball is a convex set: if $\|\mathbf{x}_1 - \mathbf{x}_0\| < r$ and $\|\mathbf{x}_2 - \mathbf{x}_0\| < r$, then for $\theta \in [0, 1]$, $\|(\theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2) - \mathbf{x}_0\| = \|\theta(\mathbf{x}_1 - \mathbf{x}_0) + (1 - \theta)(\mathbf{x}_2 - \mathbf{x}_0)\| \leq \theta \|\mathbf{x}_1 - \mathbf{x}_0\| + (1 - \theta) \|\mathbf{x}_2 - \mathbf{x}_0\| < r$.
(**Triangular Inequality**³)

³Specifically, $\forall \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$, with equality only when $\mathbf{v} = c\mathbf{w}$, where c is a constant.

Important Examples - Cont'd*

- An *ellipsoid* is related to a ball, and has the form $\epsilon(\mathbf{x}_0) = \{\mathbf{x} | (\mathbf{x} - \mathbf{x}_0)^\top \mathbf{P}^{-1} (\mathbf{x} - \mathbf{x}_0) < \epsilon\}$, where $\epsilon > 0$ and $\mathbf{P} \in \mathbb{R}^{n \times n}$ is symmetric and positive definite. The matrix $\mathbf{P}$ determines how far the ellipsoid extends in every direction from $\mathbf{x}_0$. Now we prove that an ellipsoid is convex.

Proof.

Since $\mathbf{P}$ is symmetric and positive definite, so is $\mathbf{P}^{-1}$. Then we can decompose to get $\mathbf{P}^{-1} = \mathbf{Q}^\top \mathbf{Q}$, where $\mathbf{Q}$ is some symmetric and positive definite matrix.

If $\mathbf{x}_1, \mathbf{x}_2 \in \epsilon(\mathbf{x}_0)$, then for $\theta \in [0, 1]$,

$$\begin{aligned}
 & ((\theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2) - \mathbf{x}_0)^\top \mathbf{P}^{-1} ((\theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2) - \mathbf{x}_0) \\
 &= (\mathbf{Q}(\theta(\mathbf{x}_1 - \mathbf{x}_0) + (1 - \theta)(\mathbf{x}_2 - \mathbf{x}_0)))^\top (\mathbf{Q}(\theta(\mathbf{x}_1 - \mathbf{x}_0) + (1 - \theta)(\mathbf{x}_2 - \mathbf{x}_0))) \\
 &= \theta^2 (\mathbf{Q}(\mathbf{x}_1 - \mathbf{x}_0))^\top (\mathbf{Q}(\mathbf{x}_1 - \mathbf{x}_0)) + 2\theta(1 - \theta) \underbrace{(\mathbf{Q}(\mathbf{x}_1 - \mathbf{x}_0))^\top (\mathbf{Q}(\mathbf{x}_2 - \mathbf{x}_0))}_{\text{a scalar, equal to its transpose}} + (1 - \theta)^2 (\mathbf{Q}(\mathbf{x}_2 - \mathbf{x}_0))^\top (\mathbf{Q}(\mathbf{x}_2 - \mathbf{x}_0)) \\
 &\leq \theta^2 (\mathbf{x}_1 - \mathbf{x}_0)^\top \mathbf{P}^{-1} (\mathbf{x}_1 - \mathbf{x}_0) \\
 &\quad + 2\theta(1 - \theta) \sqrt{(\mathbf{x}_1 - \mathbf{x}_0)^\top \mathbf{P}^{-1} (\mathbf{x}_1 - \mathbf{x}_0)} \sqrt{(\mathbf{x}_2 - \mathbf{x}_0)^\top \mathbf{P}^{-1} (\mathbf{x}_2 - \mathbf{x}_0)} \quad (\text{Cauchy-Schwarz Inequality}) \\
 &\quad + (1 - \theta)^2 (\mathbf{x}_2 - \mathbf{x}_0)^\top \mathbf{P}^{-1} (\mathbf{x}_2 - \mathbf{x}_0) \\
 &< \theta^2 \epsilon + 2\theta(1 - \theta) \sqrt{\epsilon} \sqrt{\epsilon} + (1 - \theta)^2 \epsilon = \epsilon.
 \end{aligned}$$

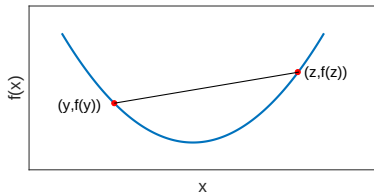
□

Cauchy-Schwarz Inequality: $\forall \mathbf{v}, \mathbf{w} \in \mathbb{R}^n, |\mathbf{v}^\top \mathbf{w}| \leq \|\mathbf{v}\| \|\mathbf{w}\|$, with equality only when $\mathbf{v} = c\mathbf{w}$, where c is a constant.

Convex Functions

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if **dom** f , which denotes the domain of f , is a convex set and if for all $\mathbf{y}, \mathbf{z} \in \mathbf{dom} f$, and $\theta \in [0, 1]$,

$$f(\theta \mathbf{y} + (1 - \theta) \mathbf{z}) \leq \theta f(\mathbf{y}) + (1 - \theta) f(\mathbf{z}). \quad (29)$$



Geometrically, this inequality means that the line segment between $(\mathbf{y}, f(\mathbf{y}))$ and $(\mathbf{z}, f(\mathbf{z}))$ lies above the graph of f between the two points.

A function f is **strictly convex** if strict inequality holds in (29) whenever $\mathbf{y} \neq \mathbf{z}$ and $\theta \in (0, 1)$.

Concave and Affine Functions

We say f is **concave** if $-f$ is convex, and **strictly concave** if $-f$ is strictly convex.

For an affine function we always have equality in (29), so **all affine (and also linear) functions are both convex and concave**. Conversely, any function that is convex and concave is affine (subject to further technical conditions).⁴

— Affine functions: $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$.

— Linear functions: $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$.

⁴Suppose a $\mathbf{C}^1$ function f is both convex and concave on a convex set $\mathcal{C}$. For all $\mathbf{y}, \mathbf{z} \in \mathcal{C}$, and $\theta \in [0, 1]$, we have $f(\theta\mathbf{y} + (1 - \theta)\mathbf{z}) \leq \theta f(\mathbf{y}) + (1 - \theta)f(\mathbf{z})$ and $f(\theta\mathbf{y} + (1 - \theta)\mathbf{z}) \geq \theta f(\mathbf{y}) + (1 - \theta)f(\mathbf{z})$. Thus, $f(\theta\mathbf{y} + (1 - \theta)\mathbf{z}) = \theta f(\mathbf{y}) + (1 - \theta)f(\mathbf{z})$. Taking derivatives with respect to θ at $\theta = 0+$, we obtain $f_\theta(\theta\mathbf{y} + (1 - \theta)\mathbf{z})|_{\theta=0+} = \lim_{\theta \rightarrow 0+} \frac{f(\theta\mathbf{y} + (1 - \theta)\mathbf{z}) - f(\mathbf{z})}{\theta} = (\nabla f(\mathbf{z}))^\top (\mathbf{y} - \mathbf{z}) = f(\mathbf{y}) - f(\mathbf{z})$. This gives $f(\mathbf{y}) = (\nabla f(\bar{\mathbf{z}}))^\top \mathbf{y} + (f(\bar{\mathbf{z}}) - (\nabla f(\bar{\mathbf{z}}))^\top \bar{\mathbf{z}})$, for all $\mathbf{y} \in \mathcal{C}$ and a fixed $\bar{\mathbf{z}}$.

Convexity of Functions

Convexity is a general concept used in optimization theory. It is responsible for the nice result that any **local** minimizer of a convex function is also a **global** minimizer. (See more later.)

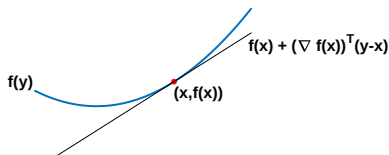
However, it is usually inconvenient to bruteforcely apply the definition to test whether a specific function is convex. For well-behaved $\mathcal{C}^1$ or $\mathcal{C}^2$ functions, i.e., those with continuous first- or second-order derivatives, fortunately, there are theorems that connect convexity with first or second derivatives.

First-Order Conditions for Convexity

Suppose f is $\mathbf{C}^1$ (e.g., its gradient ∇f exists and is continuous at each point in $\text{dom } f$, which is open). Then f is convex **if and only if** $\text{dom } f$ is convex and

$$f(\mathbf{y}) \geq f(\mathbf{x}) + (\nabla f(\mathbf{x}))^\top (\mathbf{y} - \mathbf{x}) \quad (30)$$

holds for all $\mathbf{x}, \mathbf{y} \in \text{dom } f$. (See Section 3.1.3 of Boyd and Vandenberghe (2004) for proof.)



For a convex function, the first-order Taylor approximation is a *global underestimator* of the function.

First-Order Conditions for Convexity

The inequality (30) shows that from *local information* about a convex function (i.e., its value and derivative at a point) we can derive *global information* (i.e., a global underestimator of it).

This is perhaps the most important property of convex functions, and explains some of the remarkable properties of convex functions and convex optimization problems.

As one simple example, the inequality (30) shows that if $\nabla f(\mathbf{x}) = 0$, then for all $\mathbf{y} \in \text{dom } f$, $f(\mathbf{y}) \geq f(\mathbf{x})$, i.e., $\mathbf{x}$ is a **global minimizer** of the function f .

First-Order Conditions for Strict Convexity

Similarly, a $\mathbf{C}^1$ function f is strictly convex if and only if $\mathbf{dom} f$ is convex and open, and for all $\mathbf{x}, \mathbf{y} \in \mathbf{dom} f$ and $\mathbf{x} \neq \mathbf{y}$,

$$f(\mathbf{y}) > f(\mathbf{x}) + (\nabla f(\mathbf{x}))^\top (\mathbf{y} - \mathbf{x}). \quad (31)$$

As another simple example, the inequality (31) shows that if $\nabla f(\mathbf{x}) = 0$, then for all $\mathbf{y} \in \mathbf{dom} f$ and $\mathbf{y} \neq \mathbf{x}$, $f(\mathbf{y}) > f(\mathbf{x})$, i.e., $\mathbf{x}$ is the unique global minimizer of the strictly convex function f .

Second-Order Conditions for Convexity

The first-order conditions are still somewhat cumbersome to use. Fortunately and more relevant to our purpose, we also have second-order conditions for convexity.

If f is twice differentiable at an interior point $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$ of $\mathbf{dom} f$,

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_2^2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_n} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_n^2} \end{bmatrix}$$

is called the **Hessian matrix** after German mathematician Ludwig Otto Hesse.

Suppose f is $\mathcal{C}^2$ on $\mathbf{dom} f$, which is open. Then f is convex **if and only if** $\mathbf{dom} f$ is convex and $\nabla^2 f(\mathbf{x})$ is **positive semi-definite** for all $\mathbf{x} \in \mathbf{dom} f$. And f is strictly convex **if** $\mathbf{dom} f$ is convex and $\nabla^2 f(\mathbf{x})$ is **positive definite** for all $\mathbf{x} \in \mathbf{dom} f$. (See Section 3.1.4 of Boyd and Vandenberghe (2004) for proof.)

Recap on Open Sets

For the purpose of this course, we consider sets in the Euclidean n -space, where distance is defined by the familiar Euclidean norm.

Given a set $\mathcal{S} \subseteq \mathbb{R}^n$.

- A point $\mathbf{x}$ in $\mathcal{S}$ is **interior**, if there is $B_r(\mathbf{x}) \subseteq \mathcal{S}$. That is, the point has a neighbourhood which is completely contained in $\mathcal{S}$. For example, $(0, 0)$ is an interior point of the set $\{(x_1, x_2) | x_1^2 + x_2^2 \leq 1\}$, but $(0, 1)$ is not.
- A set $\mathcal{S}$ is **open** if every point of $\mathcal{S}$ is interior. For instance, the subset $(0, 1)$ of $\mathbb{R}$ is open. The affine set $\{(x_1, x_2) | x_1 + x_2 = 1\}$ is not an open subset of $\mathbb{R}^2$, but the convex set $\{(x_1, x_2) | x_1 + x_2 < 1\}$ is. The convex set $\{(x_1, x_2) | x_1^2 + x_2^2 \leq 1\}$ is not open, but the convex set $\{(x_1, x_2) | x_1^2 + x_2^2 < 1\}$ is.
- The empty set $\emptyset$ is, by definition, open.

Recap on Open Sets

Given a set $\mathcal{S} \subseteq \mathbb{R}^n$.

- We define the **interior** of $\mathcal{S}$ (denoted $\text{int } \mathcal{S}$) by

$$\text{int } \mathcal{S} = \{\mathbf{x} \in \mathcal{S} \mid \text{there is } B_r(\mathbf{x}) \subseteq \mathcal{S}\}.$$

Clearly, $\text{int } \mathcal{S}$ is open.

The interior of $\{(x_1, x_2) \mid x_1 + x_2 \leq 1\}$ is $\{(x_1, x_2) \mid x_1 + x_2 < 1\}$. The interior of $\{(x_1, x_2) \mid x_1^2 + x_2^2 \leq 1\}$ is $\{(x_1, x_2) \mid x_1^2 + x_2^2 < 1\}$.

- We define the **boundary** of $\mathcal{S}$ (denoted $\partial \mathcal{S}$) by

$$\partial \mathcal{S} = \{\mathbf{x} \mid \text{Any } B_r(\mathbf{x}) \text{ will contain at least one point in and one point out of } \mathcal{S}\}.$$

The boundary of $\{(x_1, x_2) \mid x_1 + x_2 \leq 1\}$ is $\{(x_1, x_2) \mid x_1 + x_2 = 1\}$. The boundary of $\{(x_1, x_2) \mid x_1 + x_2 < 1\}$ is also $\{(x_1, x_2) \mid x_1 + x_2 = 1\}$.

- The boundary of the empty set $\emptyset$ is just the empty set.

“Unconstrained” versus “Constrained” Optima

We restate an optimization problem in the following.

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}), \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^n. \end{aligned} \tag{32}$$

- A point $\mathbf{x}^*$ where f achieves a minimum will be called an **unconstrained minimizer** if $\mathbf{x}^* \in \text{int } \mathcal{X}$.
- The terminology “unconstrained”, while standard, is somewhat unfortunate, since it does **not** mean the absence of constraints. Rather, it refers to the more general situation that **the constraints have no bite on the optimal point**, that is, a situation where we can move away by a small distance from the optimal point in any direction, without leaving the feasible set $\mathcal{X}$. In this sense, the constraints are **not** binding.
- If $\mathcal{X}$ is open, then any minimizer would be interior and hence unconstrained.

“Unconstrained” versus “Constrained” Optima

We restate an optimization problem in the following.

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}), \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^n. \end{aligned} \tag{33}$$

- A point $\mathbf{x}^*$ where f achieves a minimum will be called a **constrained minimizer** if $\mathbf{x}^* \in \partial\mathcal{X}$. Then we are bound to leave the feasible set if we move away in some direction(s).
- For instance, consider $\mathcal{X} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}^\top \mathbf{x} = b\}$, where $\mathbf{a} \in \mathbb{R}^n$ and $\mathbf{a} \neq \mathbf{0}$. Since $\partial\mathcal{X} = \mathcal{X}$, any minimizer is on the boundary and thus constrained.
- The feasible set $\mathcal{X}$ of a constrained minimum **cannot** be open.

Univariate Minimization

Perhaps, the **simplest** optimization problem is a **univariate unconstrained minimization problem** as follows.

$$\min_x f(x), \quad (34)$$

where $x \in \mathbb{R}$ is allowed to take any value in **dom** f , which is open.

Importance of univariate optimization:

- 1 Many **real problems** correspond to finding optima of univariate functions (e.g., optimal hedge ratios of derivatives, of course, given inputs like volatility, interest rates, contract variables, etc.).
- 2 Multivariate optimization methods in commercial use today mostly contain a **line search step**.
- 3 **Fundamental ideas are best illustrated with simple univariate cases, and they usually carry over to multivariate optimization.**

Types of Minima

Generally, **dom** f **may or may not** be open. And the feasible set $\mathcal{X}$ **may or may not** be open as well.

A **global** solution is defined as follows.

- (i) **A Global Minimum:** There exists a point $x^* \in \mathcal{X}$, s.t. $f(x^*) \leq f(x)$, $\forall x \in \mathcal{X}$.

An “**unconstrained**” **local** solution can be strong or weak.

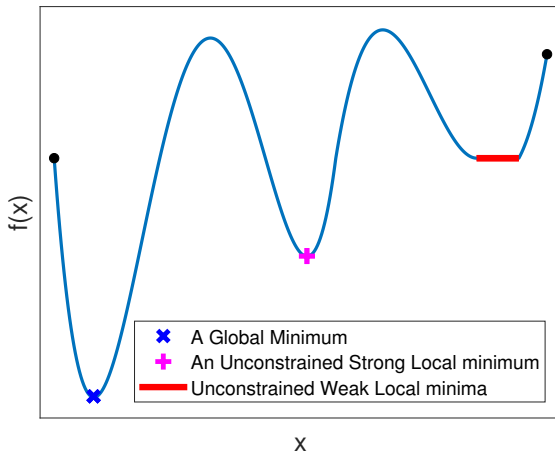
- (ii) **An Unconstrained Strong Local Minimum:** There exists a point $x^* \in \mathcal{X}$ and a deleted neighbourhood $\dot{B}_r(x^*) \subseteq \mathcal{X}$, s.t. $f(x^*) < f(x)$, $\forall x \in \dot{B}_r(x^*)$.
- (iii) **An Unconstrained Weak Local Minimum:** There exists a point $x^* \in \mathcal{X}$ and a neighbourhood $B_r(x^*) \subseteq \mathcal{X}$, s.t. $f(x^*) \leq f(x)$, $\forall x \in B_r(x^*)$.

Clearly,

$$\begin{aligned} (i) &\nRightarrow (ii), \quad (ii) \nRightarrow (i), \\ (ii) &\Rightarrow (iii), \quad (iii) \nRightarrow (ii), \\ (i) &\nRightarrow (iii), \quad (iii) \nRightarrow (i). \end{aligned}$$

However, if x^* is interior (e.g., when $\mathcal{X}$ is open), then $(i) \Rightarrow (iii)$.

An Illustration



Necessary and Sufficient Conditions for Minima

Taylor's formula turns out to be key to obtain necessary and sufficient conditions for unconstrained minimization problems.

And it naturally introduces the crucial concept of **convexity of objective functions** for **globality** of minima.

Taylor's Formula

Taylor's Formula in *Lagrange's Form*

Let $f : \mathcal{I} = (c, d) \rightarrow \mathbb{R}$ be a n -times differentiable function. If a, b are distinct points in $\mathcal{I}$, then there exists a point $\bar{x}$ strictly between a and b such that

$$\begin{aligned} f(b) &= f(a) + f'(a)(b-a) + \frac{f''(a)}{2}(b-a)^2 + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(b-a)^{n-1} \\ &\quad + \frac{f^{(n)}(\bar{x})}{n!}(b-a)^n \\ &= \sum_{i=0}^{n-1} \frac{f^{(i)}(a)}{i!}(b-a)^i + \frac{f^{(n)}(\bar{x})}{n!}(b-a)^n. \end{aligned} \tag{35}$$

Note that the case $n = 1$ is precisely the mean value theorem.

See Theorem 1.1 of Güler (2010): *Foundations of Optimization*, Springer.

Application to Optimization

In practice, the **most useful** cases of Taylor's theorem correspond to $n = 1$ and $n = 2$. Now we illustrate the use of the case $n = 2$ in optimization.

For a $\mathcal{C}^2$ function $f(x)$ on $\mathcal{I}$, suppose that x^* is a critical point, i.e., $f'(x^*) = 0$. The quadratic Taylor's formula gives

$$f(x) = f(x^*) + \frac{f''(\bar{x})}{2}(x - x^*)^2, \quad (36)$$

for some $\bar{x}$ strictly between x and x^* .

- If $f''(\bar{x}) \geq 0$ for all $\bar{x} \in \mathcal{I}$, then $f(x^*) \leq f(x)$ for all $x \in \mathcal{I}$. This shows that x^* is a **global minimizer** of f . Recall that a function f with $f''(x)$ nonnegative at all points has positive semi-definite Hessian matrices in its domain, and hence is a **convex function**.
- If $f''(\bar{x}) \geq 0$ only in a neighbourhood of x^* , then $f(x^*) \leq f(x)$ in that neighbourhood. Thus, x^* is an unconstrained weak **local minimizer** of f , but **not** necessarily a global minimizer of f .

Necessary Conditions for Minima

For a $\mathcal{C}^2$ function $f(x)$ on $\mathcal{I}$, it is obvious that a minimizer x^* must be a critical point, that is, $f'(x^*) = 0$. Or else we could move to the left or right to reduce the function value.

Necessary Conditions for an Unconstrained Weak Local Minimum

C1. $f'(x^*) = 0$ (stationarity).

C2. $f''(x^*) \geq 0$.

For C2, fix any neighbourhood $B_{r_0}(x^*)$. Suppose on the contrary, $f''(x^*) < 0$. Then we can find a sufficiently small $r_1 > 0$, such that $f''(x) < 0$ in $B_{r_1}(x^*)$, because $f''(x)$ is continuous. Equation (36) shows that x^* cannot be a minimizer in $B_{r_1 \wedge r_0}(x^*) \subseteq B_{r_0}(x^*)$, where $r_1 \wedge r_0 \equiv \min(r_1, r_0)$. Since $B_{r_0}(x^*)$ is arbitrarily, x^* cannot be a minimizer in any neighbourhood.

Sufficient Conditions for Minima

Sufficient Conditions for an Unconstrained Strong Local Minimum

C1. $f'(x^*) = 0$ (stationarity).

C2. $f''(x^*) > 0$.

For C2, $f''(x) > 0$ in a neighbourhood of x^* , because $f''(x)$ is continuous. Equation (36) shows that x^* is a strong minimizer in this neighbourhood.

Sufficient Conditions for Minima

Sufficient Conditions for an Unconstrained Weak Local Minimum

C1. $f'(x^*) = 0$ (stationarity).

C2. There exists a neighbourhood $B_r(x^*) \subseteq \mathcal{I}$, s.t. $f''(x) \geq 0$ for all $x \in B_r(x^*)$.

Sufficient Conditions for a Global Minimum

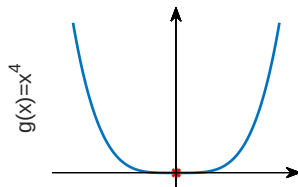
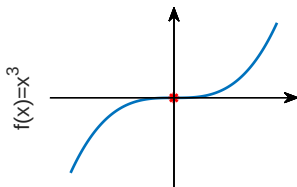
C1. $f'(x^*) = 0$ (stationarity).

C2. $f''(x) \geq 0$ for all $x \in \mathcal{I}$.

An Example

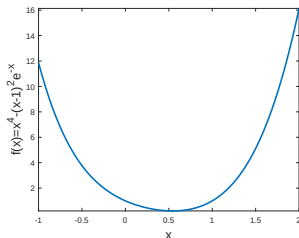
Let's consider $f(x) = x^3$ and $g(x) = x^4$, both defined on $(-\infty, +\infty)$, and $x^* = 0$.

- The condition C1 passes, because $f'(x^*) = g'(x^*) = 0$.
- For the condition C2, $f''(x^*) = 6x^* = 0$ and $g''(x^*) = 12x^{*2} = 0$. Moreover, $g''(x) = 12x^2 \geq 0$ for all x . Hence C2 passes for a global minimizer of $g(x)$. It also passes for an unconstrained weak local minimizer of $g(x)$. Note that we cannot tell whether x^* is an unconstrained strong local minimizer of $g(x)$ even though it actually is, as the sufficiency condition we supply fails. On the other hand, x^* only satisfies the necessary condition of an unconstrained weak local minimizer of $f(x)$. We cannot tell whether it is a minimizer at all.



A More Involved Example

$$\min_x \quad x^4 - (x-1)^2 e^{-x}$$



$$\text{F.O.C.: } 4(x^*)^3 - 2(x^* - 1)e^{-x^*} + (x^* - 1)^2 e^{-x^*} = 0. \Rightarrow x^* = ?$$

$$\begin{aligned} \text{S.O.C.: } & 12(x^*)^2 + 2(2x^* - 3)e^{-x^*} - (x^* - 1)^2 e^{-x^*} \\ & = 4(x^*)^3 + 12(x^*)^2 + 2(x^* - 2)e^{-x^*} \stackrel{?}{\geq} 0? \end{aligned}$$

Discussions

Three observations:

- 1 The F.O.C. may be a nonlinear equation which is often **as difficult to solve as the original optimization problem**.
- 2 The sign of higher-order derivatives may be **difficult to determine**.
- 3 In practical problems, the objective functions and their derivatives may **only be computed numerically**.

In sum, unconstrained univariate optimization problems may not be as straightforward as you thought.⁵

⁵For instance, recall the problem of finding the optimal hedge ratio of a derivative contract w.r.t. its underlying asset. In the case with time-varying volatility and event risk, the price of the derivative as a function of the underlying price is usually a solution to a partial differential equation, which promises none solution expressible by elementary functions.

Multivariate Optimization

Now we extend to **multivariate** minimization problems.

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}), \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^n. \end{aligned} \tag{37}$$

where the feasible set $\mathcal{X}$ may or may not be open.

Types of Minima

We can likewise define global and local minima of $f(\mathbf{x})$.

- (i) **A Global Minimum:** There exists a point $\mathbf{x}^* \in \mathcal{X}$, s.t. $f(\mathbf{x}^*) \leq f(\mathbf{x})$, $\forall \mathbf{x} \in \mathcal{X}$.
- (ii) **An Unconstrained Strong Local Minimum:** There exists a point $\mathbf{x}^* \in \mathcal{X}$ and a deleted neighbourhood $\dot{B}_r(\mathbf{x}^*) \subseteq \mathcal{X}$, s.t. $f(\mathbf{x}^*) < f(\mathbf{x})$, $\forall \mathbf{x} \in \dot{B}_r(\mathbf{x}^*)$.
- (iii) **An Unconstrained Weak Local Minimum:** There exists a point $\mathbf{x}^* \in \mathcal{X}$ and a neighbourhood $B_r(\mathbf{x}^*) \subseteq \mathcal{X}$, s.t. $f(\mathbf{x}^*) \leq f(\mathbf{x})$, $\forall \mathbf{x} \in B_r(\mathbf{x}^*)$.

Again,

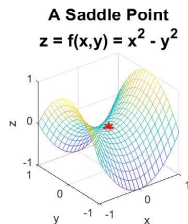
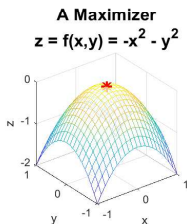
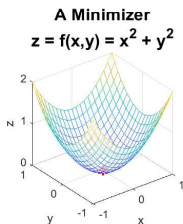
$$\begin{aligned}
 (i) &\not\Rightarrow (ii), \quad (ii) \not\Rightarrow (i), \\
 (ii) &\Rightarrow (iii), \quad (iii) \not\Rightarrow (ii), \\
 (i) &\not\Rightarrow (iii), \quad (iii) \not\Rightarrow (i).
 \end{aligned}$$

However, if $\mathbf{x}^*$ is interior (e.g., when $\mathcal{X}$ is open), then $(i) \Rightarrow (iii)$.

Stationarity

A stationary/critical point $\mathbf{x}^*$ of the function $f : \mathcal{U} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, is one satisfying $\nabla f(\mathbf{x}^*) = \mathbf{0}$.

There are three types of stationary points.



Taylor's Formula for Functions of Several Variables

Quadratic Multivariate Taylor's Formula (Theorem 1.23 of Güler (2010))

Let $\mathcal{U}$ be an open subset of $\mathbb{R}^n$, and let $\mathbf{a}, \mathbf{b}$ be two distinct points in $\mathcal{U}$ such that the line segment between $\mathbf{a}$ and $\mathbf{b}$ lies in $\mathcal{U}$. If $f : \mathcal{U} \rightarrow \mathbb{R}$ has continuous 2nd-order partial derivatives on $\mathcal{U}$, then there exists a point $\bar{\mathbf{x}}$ strictly between $\mathbf{a}$ and $\mathbf{b}$, that is, $\bar{\mathbf{x}} = \theta \mathbf{a} + (1 - \theta) \mathbf{b}$, where $0 < \theta < 1$, such that

$$f(\mathbf{b}) = f(\mathbf{a}) + (\nabla f(\mathbf{a}))^\top (\mathbf{b} - \mathbf{a}) + \frac{1}{2} (\mathbf{b} - \mathbf{a})^\top \nabla^2 f(\bar{\mathbf{x}}) (\mathbf{b} - \mathbf{a}), \quad (38)$$

Note that we can apply (38) to any two distinct points in $\mathcal{U}$ if $\mathcal{U}$ is a convex set.

Application to Optimization

For a $\mathcal{C}^2$ function $f(\mathbf{x})$ on an open and convex set $\mathcal{U}$, suppose that $\mathbf{x}^*$ is a critical point, i.e., $\nabla f(\mathbf{x}^*) = \mathbf{0}$. The quadratic Taylor's formula gives

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^\top \nabla^2 f(\bar{\mathbf{x}})(\mathbf{x} - \mathbf{x}^*), \quad (39)$$

for all $\mathbf{x}$ in $\mathcal{U}$ and for some $\bar{\mathbf{x}}$ strictly between $\mathbf{x}$ and $\mathbf{x}^*$.

- If for all $\bar{\mathbf{x}} \in \mathcal{U}$, $\nabla^2 f(\bar{\mathbf{x}})$ is positive semi-definite, that is, $\mathbf{v}^\top \nabla^2 f(\bar{\mathbf{x}}) \mathbf{v} \geq 0$ for all $\mathbf{v} \in \mathbb{R}^n$, then $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{U}$. This shows that $\mathbf{x}^*$ is a *global minimizer* of f . A function f with $\nabla^2 f(\mathbf{x})$ positive semi-definite at all points is a *convex function* (the domain $\mathcal{U}$ is convex).
- If $\nabla^2 f(\bar{\mathbf{x}})$ is positive semi-definite only in a neighbourhood of $\mathbf{x}^*$, then $\mathbf{x}^*$ is an unconstrained weak *local minimizer* of f , but **not** necessarily a global minimizer of f .

Necessary Conditions for Minima

The conditions are simple multivariate extensions of those for univariate optimization.

Necessary Conditions for an Unconstrained Weak Local Minimum

C1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$ (stationarity).

C2. $\nabla^2 f(\mathbf{x}^*)$ is positive semi-definite.

We give a brief proof of C1, which is less obvious in the multivariate case.

For any $\mathbf{d} \in \mathbb{R}^n$, the directional derivative at $\mathbf{x}^*$ along $\mathbf{d}$ is

$$f'(\mathbf{x}^*; \mathbf{d}) = \lim_{t \rightarrow 0} \frac{f(\mathbf{x}^* + t\mathbf{d}) - f(\mathbf{x}^*)}{t} = (\nabla f(\mathbf{x}^*))^\top \mathbf{d}. \quad (40)$$

Since $\mathbf{x}^*$ is an unconstrained weak local minimizer, if $|t|$ is sufficiently small, the numerator is always nonnegative. Then for $t > 0$, the quotient is

nonnegative, and $\lim_{t \rightarrow 0+} \frac{f(\mathbf{x}^* + t\mathbf{d}) - f(\mathbf{x}^*)}{t} \geq 0$. Similarly, for $t < 0$, the quotient is nonpositive, and $\lim_{t \rightarrow 0-} \frac{f(\mathbf{x}^* + t\mathbf{d}) - f(\mathbf{x}^*)}{t} \leq 0$. We conclude that

$f'(\mathbf{x}^*; \mathbf{d}) = (\nabla f(\mathbf{x}^*))^\top \mathbf{d} = 0$. Pick $\mathbf{d} = \nabla f(\mathbf{x}^*)$, then $(\nabla f(\mathbf{x}^*))^\top \nabla f(\mathbf{x}^*) = 0$.

Thus $\nabla f(\mathbf{x}^*) = \mathbf{0}$.

Necessary Conditions for Minima - Continued

The conditions are simple multivariate extensions of those for univariate optimization.

Necessary Conditions for an Unconstrained Weak Local Minimum

C1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$ (stationarity).

C2. $\nabla^2 f(\mathbf{x}^*)$ is positive semi-definite.

We introduce a nice trick of dimension-reduction to show C2.

Pick any $\mathbf{z} \in \mathbb{R}^n$. Define a real-valued function

$g(t) = f(\mathbf{x}^* + t\mathbf{z})$. Since $\mathbf{x}^*$ is an unconstrained weak local minimizer, there is a neighbourhood $B_r(\mathbf{x}^*)$ such that

$f(\mathbf{x}^*) \leq f(\mathbf{x})$, for all $\mathbf{x} \in B_r(\mathbf{x}^*)$. For a sufficiently small $\epsilon > 0$,

we have $\mathbf{x}^* + t\mathbf{z} \in B_r(\mathbf{x}^*)$ for all $t \in (-\epsilon, \epsilon)$. Then $g(t)$ is

minimized at $t^* = 0$ on $(-\epsilon, \epsilon)$. Using the necessary conditions

for an unconstrained weak local minimum of a $\mathcal{C}^2$ univariate function, we know $g''(0) = \mathbf{z}^\top \nabla^2 f(\mathbf{x}^*) \mathbf{z} \geq 0$.

Sufficient Conditions for Minima

Sufficient Conditions for an Unconstrained Strong Local Minimum

C1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$ (stationarity).

C2. $\nabla^2 f(\mathbf{x}^*)$ is positive definite. That is, $\mathbf{v}^\top \nabla^2 f(\mathbf{x}^*) \mathbf{v} > 0$ for all $\mathbf{v} \in \mathbb{R}^n$ and $\mathbf{v} \neq \mathbf{0}$.

Sufficient Conditions for Minima

“Continuity” of Positive Definiteness (Corollary 1.60 of Sundaram (1996))

For a $\mathbf{C}^2$ function $f(\mathbf{x})$, if at some point $\tilde{\mathbf{y}}$, $\nabla^2 f(\tilde{\mathbf{y}})$ is positive definite, then there is a neighbourhood $B_r(\tilde{\mathbf{y}})$ such that for all $\mathbf{y} \in B_r(\tilde{\mathbf{y}})$, $\nabla^2 f(\mathbf{y})$ is positive definite.

For $C2$, the above corollary confirms that there is a neighbourhood $B_r(\mathbf{x}^*)$ such that for all $\bar{\mathbf{x}} \in B_r(\mathbf{x}^*)$, $\nabla^2 f(\bar{\mathbf{x}})$ is positive definite. Equation (39) yields that $\mathbf{x}^*$ is a strong minimizer in this neighbourhood.

Sufficient Conditions for Minima

Sufficient Conditions for an Unconstrained Weak Local Minimum

C1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$ (stationarity).

C2. There exists a neighbourhood $B_r(\mathbf{x}^*) \subseteq \mathcal{U}$, s.t. $\nabla^2 f(\mathbf{x})$ is positive semi-definite for all $\mathbf{x} \in B_r(\mathbf{x}^*)$.

Sufficient Conditions for a Global Minimum

C1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$ (stationarity).

C2. $\nabla^2 f(\mathbf{x})$ is positive semi-definite for all $\mathbf{x} \in \mathcal{U}$.

(Try proving the three sufficient conditions by two different methods: the quadratic Taylor's formula and the first order condition for convexity.)

The Recipe

- 1 Correctly **formulate** the optimization problem.
- 2 Find a point $\mathbf{x}^*$ that could **potentially** be a solution, i.e., one satisfying the necessary conditions.
- 3 Verify this point $\mathbf{x}^*$ is **certainly** a solution, i.e., one satisfying the sufficient conditions.

An Example

Consider the point $(x^*, y^*) = (0, 0)$. The following functions are defined on $\mathbb{R}^2$.

- 1 For $f(x, y) = x^2 + y^2$, $\nabla f(x^*, y^*) = (0, 0)^\top$, and $\nabla^2 f(x, y) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ is positive definite for any (x, y) . Therefore, (x^*, y^*) is a global minimizer.
- 2 For $f(x, y) = -x^2 - y^2$, since (x^*, y^*) is a global minimizer of $-f$, it is a global maximizer of f .
- 3 For $f(x, y) = x^2 - y^2$, $\nabla f(x^*, y^*) = (0, 0)^\top$, and $\nabla^2 f(x^*, y^*) = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$ is not positive semi-definite. Therefore, (x^*, y^*) violates the necessary conditions and is not a minimizer of any kind for f . Moreover, $\nabla(-f(x^*, y^*)) = (0, 0)^\top$, and $\nabla^2(-f(x^*, y^*)) = \begin{bmatrix} -2 & 0 \\ 0 & 2 \end{bmatrix}$ is not positive semi-definite, either. Therefore, (x^*, y^*) violates the necessary conditions and is not a minimizer of any kind for $-f$. In sum, (x^*, y^*) is neither a minimizer nor a maximizer of any kind for f .

Sufficient Conditions for Saddle Points

Sufficient Conditions for a Saddle Point (Theorem 2.16 of Güler (2010))

C1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$ (stationarity).

C2. $\nabla^2 f(\mathbf{x}^*)$ is indefinite, that is, it has at least one positive and one negative eigenvalue.

Proof*

Suppose $\lambda_1 > 0$ is an eigenvalue of $\nabla^2 f(\mathbf{x}^*)$ with a corresponding eigenvector $\mathbf{d}_1 \in \mathbb{R}^n$, $\mathbf{d}_1^\top \mathbf{d}_1 = 1$. Then applying the Taylor's formula in Landau's little "oh" notation,⁶ for sufficiently small $t > 0$,

$$\begin{aligned} f(\mathbf{x}^* + t\mathbf{d}_1) &= f(\mathbf{x}^*) + \frac{t^2}{2} \mathbf{d}_1^\top \nabla^2 f(\mathbf{x}^*) \mathbf{d}_1 + o(t^2) \\ &= f(\mathbf{x}^*) + \frac{t^2}{2} \lambda_1 \mathbf{d}_1^\top \mathbf{d}_1 + o(t^2) \\ &= f(\mathbf{x}^*) + \frac{t^2}{2} \lambda_1 + o(t^2) > f(\mathbf{x}^*). \end{aligned}$$

Similarly, suppose $\lambda_2 < 0$ is another eigenvalue of $\nabla^2 f(\mathbf{x}^*)$ with a corresponding eigenvector $\mathbf{d}_2 \in \mathbb{R}^n$, $\mathbf{d}_2^\top \mathbf{d}_2 = 1$. Then for sufficiently small $t > 0$, $f(\mathbf{x}^* + t\mathbf{d}_2) < f(\mathbf{x}^*)$.

On balance, $\mathbf{x}^*$ is a saddle point.

⁶This is a direct implication of the Lagrangian form. Check it by yourself.

An Example

Consider the point $(x^*, y^*) = (0, 0)$.

For $f(x, y) = x^2 - y^2$, $\nabla f(x^*, y^*) = (0, 0)^\top$, and

$\nabla^2 f(x^*, y^*) = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$. We see that the Hessian matrix has a positive eigenvalue 2 and a negative eigenvalue -2.

Therefore, (x^*, y^*) is a saddle point of f .

The Subtle Case

Consider the following case:

- For a $\mathcal{C}^2$ function $f(\mathbf{x})$ on an open and convex set $\mathcal{U}$, suppose that $\mathbf{x}^*$ is a critical point, i.e., $\nabla f(\mathbf{x}^*) = \mathbf{0}$, and $\nabla^2 f(\mathbf{x}^*)$ is positive semi-definite but not positive definite.

This case does not violate the necessary conditions for an unconstrained weak local minimum, but it does not satisfy the sufficient conditions for a global or an unconstrained strong/weak local minimum, either. **We have to turn to other knowledge to determine whether $\mathbf{x}^*$ is a minimizer**, e.g., whether f is convex on a neighbourhood of $\mathcal{U}$ or on $\mathcal{U}$. Recall the univariate example we gave before (i.e., $f(x) = x^3$ v.s. $g(x) = x^4$ at $x^* = 0$).

Furthermore, **semi-definiteness is much harder to verify than definiteness**, as the former requires checking the principal minors of all the matrices from the permutation of the index. (See Section 1.5.2 of Sundaram (1996): *A First Course in Optimization Theory*, Cambridge University Press.)

An Example of Using “Other Knowledge”

Finally, how to deal with $x^* = 0$ for $f(x) = x^3$?

For this function, we can use the cubic Taylor's formula in little “oh” notation. For sufficiently small $t > 0$, since $f'(x^*) = f''(x^*) = 0$ and $f'''(x^*) = 6$,

$$\begin{aligned} f(x^* + t) &= f(x^*) + \frac{f'''(x^*)}{3!}t^3 + o(t^3) \\ &= f(x^*) + t^3 + o(t^3) > f(x^*). \end{aligned}$$

Similarly, for sufficiently small $(-t) > 0$,

$$f(x^* + t) = f(x^*) + t^3 + o(t^3) < f(x^*).$$

Therefore, x^* is a saddle point.

Question 2

How about $f(x) = x^n$, $n > 4$, at $x^* = 0$?

Global Optimality and Uniqueness of Solutions

We may ask two further questions:

- 1 When is a **local minimizer** **global**?
- 2 When is a global minimizer **unique**?

Answers to the two questions, again, involve the convexity concept we have familiarized ourselves with.

Further Discussions on Convexity and Optimization

Now we provide our main results regarding global optimality and uniqueness of solutions.

Global Optimality and Uniqueness of Solutions

Suppose $\mathcal{D} \subseteq \mathbb{R}^n$ is convex and contains at least two distinct points, then,

- 1 if $f : \mathcal{D} \rightarrow \mathbb{R}$ is convex, any local minimizer of f is a global minimizer of f .
- 2 if $f : \mathcal{D} \rightarrow \mathbb{R}$ is strictly convex, the set $\arg \min \{f(\mathbf{x}) | \mathbf{x} \in \mathcal{D}\}$ of minimizers of f on $\mathcal{D}$ is either empty or contains a single point.

Note two things:

- 1 We do **not** require differentiability of any kind.
- 2 We do **not** require $\mathcal{D}$ to be open. Therefore a minimizer may be an interior point of $\mathcal{D}$ or lie on the boundary of $\mathcal{D}$.

Proofs*

Part 1: Suppose $\mathbf{x}^*$ is a local minimizer but not a global one. Then there is $r > 0$, such that $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in B_r(\mathbf{x}^*) \cap \mathcal{D}$. There is also $\mathbf{y} \in \mathcal{D}$ such that $f(\mathbf{x}^*) > f(\mathbf{y})$. Let $\mathbf{z} = \theta\mathbf{x}^* + (1 - \theta)\mathbf{y}$, where $\theta \in (0, 1)$. For θ sufficiently close to 1, $\mathbf{z} \in B_r(\mathbf{x}^*) \cap \mathcal{D}$.⁷ Moreover,

$$f(\mathbf{z}) \leq \theta f(\mathbf{x}^*) + (1 - \theta)f(\mathbf{y}) < f(\mathbf{x}^*).$$

This contradicts that $\mathbf{x}^*$ is a minimizer on $B_r(\mathbf{x}^*) \cap \mathcal{D}$.

Part 2: Suppose both $\mathbf{x}_1^*$ and $\mathbf{x}_2^*$ are minimizers of f on $\mathcal{D}$. Then $f(\mathbf{x}_1^*) = f(\mathbf{x}_2^*)$. For $\theta \in (0, 1)$,

$$f(\theta\mathbf{x}_1^* + (1 - \theta)\mathbf{x}_2^*) < \theta f(\mathbf{x}_1^*) + (1 - \theta)f(\mathbf{x}_2^*) = f(\mathbf{x}_1^*).$$

This contradicts that $\mathbf{x}_1^*$ is a minimizer on $\mathcal{D}$.

⁷Since both $\mathbf{x}^*$ and $\mathbf{y}$ are in $\mathcal{D}$ which is convex, $\mathbf{z} \in \mathcal{D}$ because $\mathbf{z}$ is a point on the line segment between $\mathbf{x}^*$ and $\mathbf{y}$. For θ sufficiently close to 1, $(\mathbf{z} - \mathbf{x}^*)^\top (\mathbf{z} - \mathbf{x}^*) \leq r^2$ and hence $\mathbf{z} \in B_r(\mathbf{x}^*)$. In sum, $\mathbf{z} \in B_r(\mathbf{x}^*) \cap \mathcal{D}$.

Unconstrained Optimization Again

Go back to the unconstrained optimization problem of a $\mathcal{C}^2$ function $f(\mathbf{x})$ on an open and convex set $\mathcal{U} \subseteq \mathbb{R}^n$.

If the Hessian matrix of f is **positive definite** at any point $\mathbf{x} \in \mathcal{U}$, f is **strictly convex** on $\mathcal{U}$. Then a critical point x^* is not only an **unconstrained strong local** minimizer, but also the **unique global** minimizer.

Constrained Optimization

A **constrained** optimization problem can be written as follows.

$$\begin{array}{ll} \min_{\mathbf{x}} & f(\mathbf{x}), \\ \text{s.t.} & \mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^n. \end{array} \quad (41)$$

where the feasible set $\mathcal{X}$ is **not** open and a constrained minimizer $\mathbf{x}^*$ is in $\partial\mathcal{X} \cap \mathcal{X}$.

Types of Solutions

A **global** solution is defined as follows.

- (i) **A Global Minimum:** There exists a point $\mathbf{x}^* \in \mathcal{X}$, s.t. $f(\mathbf{x}^*) \leq f(\mathbf{x})$, $\forall \mathbf{x} \in \mathcal{X}$.

A **“constrained” local** solution is defined as follows.

- (ii) **A Constrained Local Minimum:** There exists a point $\mathbf{x}^* \in \partial\mathcal{X} \cap \mathcal{X}$ and a neighbourhood $B_r(\mathbf{x}^*)$, s.t. $f(\mathbf{x}^*) \leq f(\mathbf{x})$, $\forall \mathbf{x} \in B_r(\mathbf{x}^*) \cap \mathcal{X}$.

Clearly,

$$(i) \not\Rightarrow (ii), \quad (ii) \not\Rightarrow (i).$$

However, if $\mathbf{x}^* \in \partial\mathcal{X} \cap \mathcal{X}$, then $(i) \Rightarrow (ii)$.

The Lagrangian Method

The method of Lagrange multipliers is a strategy for finding the constrained local *minima* of a function subject to *equality* constraints.

This Lagrangian (or Lagrange) method converts a constrained optimization problem into an unconstrained one to which the derivatives tests are then applied.

- 1 Identify the stationary points from the first-order necessary conditions.
- 2 Determine whether the stationary points are minima and their types if they are, e.g., through convexity tests based on second-order conditions.

The Lagrange Theorem

Consider the optimization problem of minimizing the function $f(\mathbf{x})$ on an **open** set $\mathcal{D} \subseteq \mathbb{R}^n$, subject to $m(< n)$ equality constraints.

$$\min_{\mathbf{x}} f(\mathbf{x}), \quad (42)$$

$$\text{s.t. } g_j(\mathbf{x}) = 0, \quad j = 1, 2, \dots, m, \quad (43)$$

where $f(\mathbf{x})$ and $g_j(\mathbf{x})$ are $\mathbf{C}^1$.

If $\mathbf{x}^*$ is a minimizer and $\text{rank}(\nabla \mathbf{g}(\mathbf{x}^*)) = m$, then there are scalars $\lambda_1, \lambda_2, \dots, \lambda_m$, also called Lagrange Multipliers, such that

$$\nabla f(\mathbf{x}^*) = \sum_{j=1}^m \lambda_j \nabla g_j(\mathbf{x}^*) \iff \nabla f(\mathbf{x}^*) - \sum_{j=1}^m \lambda_j \nabla g_j(\mathbf{x}^*) = \mathbf{0}. \quad (44)$$

The gradient of the objective function is a linear combination of the gradients of the constraint functions.

We call the condition $\text{rank}(\nabla \mathbf{g}(\mathbf{x}^*)) = m$ **constraint qualification**.

Intuition

Consider a slightly simpler case of minimizing $f(\mathbf{x})$ with $n = 3$, subject to a single equality constraint $g(\mathbf{x}) = 0$. The Lagrange Theorem says that at a minimizer $\mathbf{x}^*$,

$$\nabla f(\mathbf{x}^*) = \lambda \nabla g(\mathbf{x}^*).$$

So,

- 1 Why should the gradient of the objective function be proportional to that of the constraint function? Why are the two gradients related at all?
- 2 Where does λ come from?

Intuition: For the Case of a Single Constraint

- 1 The set of points satisfying $g(\mathbf{x}) = 0$ is a *surface* in 3 dimensions, or a (maybe oddly shaped) *balloon*.
- 2 The set of points satisfying $f(\mathbf{x}) = k$ is another surface in 3 dimensions, or another (maybe oddly shaped) balloon.
- 3 Take k to be a **very small** number, e.g., smaller than the minimum of $f(\mathbf{x})$ under the constraint (suppose the minimum exists).⁸ Then the Balloon $f(\mathbf{x}) = k$ lies **entirely outside/inside** the balloon $g(\mathbf{x}) = 0$.⁹

⁸Note that k should not be strictly smaller than the **global** minimum for the corresponding **unconstrained** minimization problem (suppose the minimum exists).

⁹For the degenerate case where the **global** minimizer of the corresponding **unconstrained** problem is in the constraint set $g(\mathbf{x}) = 0$, any feasible k is larger than or equal to the global minimum. Hence, for any k , the two balloons will be tangent to or intersect with each other, but not be separated from each other.

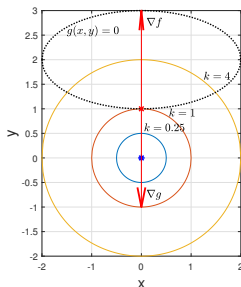
Intuition: For the Case of a Single Constraint

- 4 Now gradually increase k , by inflating air into the balloon $f(\mathbf{x}) = k$. At some point, this balloon will touch the balloon $g(\mathbf{x}) = 0$ at some point, which is the minimizer.
- 5 At the touching point of the minimizer, the two balloons should be tangent, hence their normal vectors, given by their gradients, should both be perpendicular to the same tangent plane thus parallel to each other.
- 6 Recall the Geometrical interpretation of a gradient: The direction of the gradient is the direction of fastest increase of the function at a certain point, and its magnitude is the rate of increase in that direction.
- 7 Hence, in contrast to the direction of increase, $f(\mathbf{x})$ and $g(\mathbf{x})$ need not have the same rate of increase since the two balloons may have different shapes. The scaling constant λ results.

This explanation works for any n if you are ready to think of *hyper*-balloons.

A Pictorial Illustration

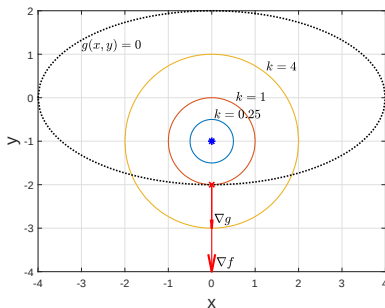
Let $n = 2$. Consider $f(x, y) = x^2 + y^2$ and $g(x, y) = x^2/4 + (y - 2)^2 - 1 = 0$. The level curves of f , defined by $x^2 + y^2 = k$, are circles. The constraint set is an ellipse.



The global minimizer of the corresponding unconstrained problem is marked by “*”, which is **outside** the bolloon $g(x, y) = 0$.

Another Pictorial Illustration

Let $n = 2$. Consider $f(x, y) = x^2 + (y + 1)^2$ and $g(x, y) = x^2/16 + y^2/4 - 1 = 0$. The level curves of f , defined by $x^2 + (y + 1)^2 = k$, are circles. The constraint set is an ellipse.



The global minimizer of the corresponding unconstrained problem is marked by “*”, which is **inside** the balloon $g(x, y) = 0$.

Intuition: For the Case of Multiple Constraints

Once we understood the case with one constraint, extension to multiple constraints is straightforward.

Consider the general case of minimizing $f(\mathbf{x})$ on $\mathbb{R}^n$, subject to $m(< n)$ equality constraints $g_j(\mathbf{x}) = 0, j = 1, 2, \dots, m$.

- 1 The Lagrange Theorem says that at a minimizer $\mathbf{x}^*$,
 $\nabla f(\mathbf{x}^*) = \sum_{j=1}^m \lambda_j \nabla g_j(\mathbf{x}^*)$.
- 2 This tells us that any direction of change, say, $\mathbf{d} \in \mathbb{R}^n$, $\mathbf{d} \neq \mathbf{0}$, that is perpendicular to **all** the gradients ∇g_j must be perpendicular to the gradient ∇f as well. That is, $(\nabla g_j(\mathbf{x}^*))^\top \mathbf{d} = 0, j = 1, 2, \dots, m. \Rightarrow$
 $(\nabla f(\mathbf{x}^*))^\top \mathbf{d} = \sum_{j=1}^m \lambda_j (\nabla g_j(\mathbf{x}^*))^\top \mathbf{d} = 0$.
- 3 This in turn means that you **cannot** decrease f without violating at least one of the constraints.

$$\begin{aligned}
 f(\mathbf{x}^* + t\mathbf{d}) - f(\mathbf{x}^*) &= t(\nabla f(\mathbf{x}^*))^\top \mathbf{d} + \frac{t^2}{2} \mathbf{d}^\top \nabla^2 f(\bar{\mathbf{x}}) \mathbf{d} \\
 &= t \underbrace{\sum_{j=1}^m \lambda_j (\nabla g_j(\mathbf{x}^*))^\top \mathbf{d}}_{< 0?!} + \underbrace{\frac{t^2}{2} \mathbf{d}^\top \nabla^2 f(\bar{\mathbf{x}}) \mathbf{d}}_{\geq 0, \text{ by sufficiency}}.
 \end{aligned}$$

The Lagrange Method

We can summary everything by defining a single Lagrangian

$$\mathcal{L}(x_1, x_2, \dots, x_n; \lambda_1, \lambda_2, \dots, \lambda_n) = f(x_1, x_2, \dots, x_n) - \sum_{j=1}^m \lambda_j g_j(x_1, x_2, \dots, x_n).$$

F.O.C.s:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial f}{\partial x_i} - \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i} = 0, \quad i = 1, 2, \dots, n,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_j} = -g_j(x_1, x_2, \dots, x_n) = 0, \quad j = 1, 2, \dots, m.$$

In one shot, we obtain $n + m$ equations, the first n from the Lagrange Theorem and the next m for constraints, with $n + m$ unknowns.

Sufficient Conditions

Now we have identified the stationary points, candidates of local minimizers. How could we verify that they are genuine minimizers?

Naturally, we may wonder whether the argument with the Hessian of an unconstrained optimization problem could be transplanted here. After all, we have made great efforts to transform a constrained optimization into an unconstrained one.

The answer is “Yes” and “No”.

- o “Yes”: We can basically apply a Hessian-based approach to perform such a job.
- o “No”: Now the optimizers must satisfy the constraints that dictate that for each j , $\nabla g_j \cdot (dx_1, dx_2, \dots, dx_n)^\top = 0$. Hence, $(dx_1, dx_2, \dots, dx_n)$ cannot be arbitrary but is confined by the constraints. We only need to check the optimality of f along the directions perpendicular to all the gradients of the constraints.

Our Approach

We relegate a detailed discussion of the so-called bordered-Hessian method for constrained optimization problems to your future study.¹⁰

We again use convexity to check sufficiency, globality, and uniqueness of constrained minimizers.

We will provide a duality approach for the mean-variance problem in Case I. As a byproduct, we derive the Lagrangian first-order condition and ensure the sufficiency of the solution for the mean-variance problem. We will also briefly talk about inequality constraints there.

¹⁰For self-study, see, for instance, Section 5.3 of Sundaram (1996).

Convexity and Constrained Optimization

$$\begin{array}{ll} \min_{\mathbf{x}} & f(\mathbf{x}), \\ \text{s.t.} & \mathbf{x} \in \mathcal{X}. \end{array} \quad (45)$$

Suppose $\text{dom } f = \mathbb{R}^n$. When the equality constraint functions are all affine, the feasible set $\mathcal{X} = \text{dom } f \cap \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{A}^\top \mathbf{x} = \mathbf{b}\}$ is the intersection of hyperplanes and thus affine (hence, convex), but **not** open. Actually, all feasible points are on the boundary. That is, $\partial \mathcal{X} = \mathcal{X}$ and $\text{int } \mathcal{X} = \emptyset$. In this sense, any minimizer will be “constrained”.

Moreover, if f is strictly convex on $\mathcal{X}$, and if there is a minimizer to the constrained optimization problem, then the minimizer is the unique global minimizer.

The Markowitz Problem

$$\begin{aligned}
 \min_{w_1, w_2, \dots, w_N} \quad & \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N w_i \sigma_{ij} w_j, \\
 \text{s.t.} \quad & \sum_{i=1}^N w_i \bar{r}_i = \bar{r}_p, \\
 & \sum_{i=1}^N w_i = 1.
 \end{aligned}$$

The objective function is **strictly convex** since its Hessian Matrix $[\sigma_{ij}]$ is **positive definite** on $\mathbb{R}^n$, where the positive definiteness follows from the fact that the variance of any portfolio is strictly positive as long as there are **no redundant securities**. And the feasible set $\mathcal{X}$ carved by the two equality constraints is a convex set.

The constraint qualification requires that all the expected returns of the individual assets should not be equal. That is, we should not have $\bar{r}_1 = \bar{r}_2 = \dots = \bar{r}_N$.

If the Lagrangian first-order necessary conditions are also sufficient, then they **identify the unique global minimizer**. We can focus on the first-order conditions for this problem and need not worry about the second-order conditions. We next turn to the issue of sufficiency under the strengthened assumption of convexity of the objective function.

Convexity and Sufficiency

Now we provide a case where the Lagrangian first-order necessary conditions are also sufficient.

Convexity and Optimization with Affine Equality Constraints

For the optimization problem considered in the Lagrange Theorem, we further assume that the objective function $f(\mathbf{x})$ is convex on an open and convex set $\mathcal{D} \subseteq \mathbb{R}^n$, and that the equality constraint functions $g_j(\mathbf{x})$, $j = 1, 2, \dots, m$, are all affine. Then, a point $\mathbf{x}^*$ satisfying $\nabla f(\mathbf{x}^*) = \sum_{j=1}^m \lambda_j \nabla g_j(\mathbf{x}^*)$ is a global minimizer.

Proof*

The feasible set is $\mathcal{X} = \mathcal{D} \cap \{\mathbf{x} \in \mathbb{R}^n | \mathbf{g}(\mathbf{x}) = 0\}$. Since $\mathcal{X}$ is the intersection of two convex sets, it is also convex. Given a point $\mathbf{x}^* \in \mathcal{X}$ and $\nabla f(\mathbf{x}^*) = \sum_{j=1}^m \lambda_j \nabla g_j(\mathbf{x}^*)$. Obviously, $g_j(\mathbf{x}^*) = 0, j = 1, 2, \dots, m$. For an arbitrary point $\mathbf{y} \in \mathcal{X}$, since $f(\mathbf{x})$ is convex on $\mathcal{D}$ and $\mathcal{X} \subseteq \mathcal{D}$, we have

$$f(\mathbf{y}) - f(\mathbf{x}^*) \geq (\nabla f(\mathbf{x}^*))^\top (\mathbf{y} - \mathbf{x}^*) = \sum_{j=1}^m \lambda_j (\nabla g_j(\mathbf{x}^*))^\top (\mathbf{y} - \mathbf{x}^*) = \sum_{j=1}^m \lambda_j \nabla g_j(\mathbf{x}^*, \mathbf{y} - \mathbf{x}^*),$$

where $\nabla g_j(\mathbf{x}^*, \mathbf{y} - \mathbf{x}^*)$ denotes the directional derivative of g_j at $\mathbf{x}^*$ along $\mathbf{y} - \mathbf{x}^*$. Since $g_j, j = 1, 2, \dots, m$, is $\mathbf{C}^1$ and thus differentiable, all directional derivatives exist and $\nabla g_j(\mathbf{x}^*, \mathbf{y} - \mathbf{x}^*) = (\nabla g_j(\mathbf{x}^*))^\top (\mathbf{y} - \mathbf{x}^*)$.

According to the definition of directional derivatives, for $j = 1, 2, \dots, m$,

$$\nabla g_j(\mathbf{x}^*, \mathbf{y} - \mathbf{x}^*) = \lim_{t \rightarrow 0} \frac{g_j(\mathbf{x}^* + t(\mathbf{y} - \mathbf{x}^*)) - g_j(\mathbf{x}^*)}{t} = \lim_{t \rightarrow 0+} \frac{g_j((1-t)\mathbf{x}^* + t\mathbf{y})}{t}.$$

For $t \in (0, 1)$, $(1-t)\mathbf{x}^* + t\mathbf{y} \in \mathcal{X}$ and thus $g_j((1-t)\mathbf{x}^* + t\mathbf{y}) = 0, j = 1, 2, \dots, m$. Furthermore,

$$\nabla g_j(\mathbf{x}^*, \mathbf{y} - \mathbf{x}^*) = \lim_{t \rightarrow 0+} 0 = 0, \quad j = 1, 2, \dots, m.$$

Finally,

$$f(\mathbf{y}) - f(\mathbf{x}^*) \geq \sum_{j=1}^m \lambda_j \nabla g_j(\mathbf{x}^*, \mathbf{y} - \mathbf{x}^*) = 0.$$

The Markowitz Problem

Now go back to the Markowitz mean-variance problem.

$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N w_i \sigma_{ij} w_j - \lambda \left(\sum_{i=1}^N w_i \bar{r}_i - \bar{r}_p \right) - \zeta \left(\sum_{i=1}^N w_i - 1 \right).$$

The F.O.C.s are

$$\sum_{j=1}^N \sigma_{ij} w_j^* - \lambda^* \bar{r}_i - \zeta^* = 0, \text{ for } i = 1, 2, \dots, N,$$

$$\sum_{i=1}^N w_i^* \bar{r}_i = \bar{r}_p,$$

$$\sum_{i=1}^N w_i^* = 1.$$

Two-Fund Separation

Note that the F.O.C.s are all **affine** functions of $(w_1, w_2, w_3, \lambda, \zeta)$. If $x^1 = (w_1^1, w_2^1, w_3^1, \lambda^1, \zeta^1)$ is a known solution for $\bar{r}_p^1$, and $x^2 = (w_1^2, w_2^2, w_3^2, \lambda^2, \zeta^2)$ is a known solution for $\bar{r}_p^2$, then, $\forall \alpha$, $\alpha x^1 + (1 - \alpha)x^2$ is a solution for $\alpha \bar{r}_p^1 + (1 - \alpha)\bar{r}_p^2$. (Check this yourself.)

The Two-Fund Theorem

Once two efficient funds (minimum variance portfolios) are established, investors seeking efficient investment in funds, in terms of mean-variance, can duplicate any efficient portfolio by simply combining the two.

It is also referred to as the **Mutual Fund Separation Theorem**. Investors need not purchase individual securities. Two mutual funds would be a complete service.

Two-Fund Separation - Discussions

The Two-Fund Theorem relies crucially on the following assumptions.

- 1 Investors only care mean and variance.
- 2 Investors have the same assessment of the means, variances and covariances.
- 3 A single-period investment horizon.

All the three assumptions are tenuous in reality. But **the theorem provides a good way to understand the investment process.**

- o For instance, if investors care more than mean and variance and invest for multiple periods, the two-fund separation is no long attained. However, we may have a **three-fund separation**.¹¹

¹¹See Merton, Robert C., 1973. An intertemporal capital asset pricing model, *Econometrica* 41(5), 867-887.

Adding a Riskfree Asset

So far we have focused on risky assets with $\sigma_i > 0$.

Introduction of a riskfree asset with $\sigma_{r_f} = 0$ enables borrowing and lending at the riskfree rate r_f .

Perhaps surprisingly, adding one more riskfree asset causes a mathematical degeneracy hence greatly simplifies the shape of the efficient frontier.

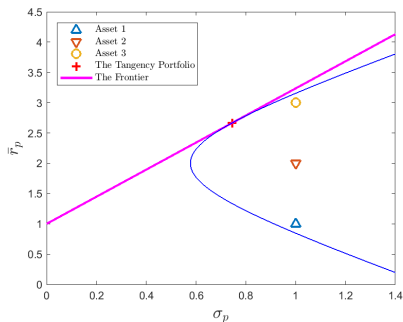
For a portfolio with a weight α ($\alpha \leq 1$) in the riskfree asset and a weight $1 - \alpha$ in a risky asset with mean return of $\bar{r}$ and standard deviation σ ,

$$\bar{r}_p = \alpha r_f + (1 - \alpha)\bar{r}, \quad \sigma_p = (1 - \alpha)\sigma.$$

The Efficient Frontier

Let $r_f = 1$. The frontier with three uncorrelated risky assets but without the riskfree asset, the case we have studied, is described by $\sigma_p = \sqrt{\bar{r}_p^2/2 - 2\bar{r}_p + 7/3}$.

Now we turn to calculate the weights of the tangency portfolio.



Question 3

On the Tangency Portfolio

What are the weights of the tangency portfolio?

Question 3 - Answer

On the Tangency Portfolio

What are the weights of the tangency portfolio?

The frontier of risky assets is the curve $\sigma_p^2 = \bar{r}_p^2/2 - 2\bar{r}_p + 7/3$.

Differentiate both sides, we obtain:

$$2\sigma_p d\sigma_p = \bar{r}_p d\bar{r}_p - 2d\bar{r}_p, \quad \implies \quad \frac{d\bar{r}_p}{d\sigma_p} = \frac{2\sigma_p}{\bar{r}_p - 2}$$

The slope of the tangency line is $\frac{\bar{r}_p - r_f}{\sigma_p}$. Let the two slopes equal.

$$\implies \quad \bar{r}_p = \frac{14/3 - 2r_f}{2 - r_f}.$$

Then, $w_1^* = 1/3 - (\bar{r}_p/2 - 1)$, $w_2^* = 1/3$, and $w_3^* = 1/3 + (\bar{r}_p/2 - 1)$. Putting in $r_f = 1$, we get $\bar{r}_p = 8/3$ and $(w_1^*, w_2^*, w_3^*) = (0, 1/3, 2/3)$.

The One-Fund Theorem

Any efficient portfolio is a combination of riskfree borrowing/lending and a single fund of risky assets.

Note that on the aggregate level, borrowing and lending cancel out. The weights would be 100% in the risky fund, and 0% in the riskfree asset.

Case I